

14

Time series 2

Syllabus objectives

- 2.1 Concepts underlying time series models
 - 2.1.3 Explain the concept of a filter applied to a stationary random series.
 - 2.1.7 Explain the basic concept of a multivariate autoregressive model.
 - 2.1.8 Explain the concept of cointegrated time series.
- 2.2 Applications of time series models
 - 2.2.1 Outline the process of identification, estimation and diagnosis of a time series, the criteria for choosing between models and the diagnostic tests that might be applied to the residuals of a time series after estimation.
 - 2.2.2 Describe briefly other non-stationary, non-linear time series models.
 - 2.2.3 Describe simple applications of a time series model, including random walk, autoregressive and cointegrated models as applied to security prices and other economic variables.
 - 2.2.4 Develop deterministic forecasts from time series data, using simple extrapolation and moving average models, applying smoothing techniques and seasonal adjustment when appropriate.

0 Introduction

We have seen in [Chapter 13](#) that stationary time series are easier to analyse than non-stationary ones. In this chapter we look more closely at the causes of non-stationarity, and we consider a number of related questions:

1. How can we tell whether a time series is stationary or not?
2. If it is not, how do we go about turning it into a stationary series?

We shall use the characteristics of the different types of time series that we studied in [Chapter 13](#) to help to determine an appropriate approach. The ACF and PACF will be key tools here.

We also look at the Box-Jenkins approach to fitting and forecasting.

Once we have chosen an appropriate time series model, two further questions arise.

1. How do we use the model to determine estimates of the future values of our time series?
2. Is the model we have chosen a good one? How well does it fit the data that we already have?

We shall look at possible answers to both these questions in this chapter.

1 Compensating for trend and seasonality

All the methods which we shall investigate in Sections 3 and 4 apply only to a time series which gives the appearance of stationarity. In this section, therefore, we deal with possible sources of non-stationarity and how to compensate for them.



A simple time series plot in R can be generated as:

```
ts.plot(x)
```

where x is some (vector) time series data.

Lack of stationarity may be caused by the presence of deterministic effects in the quantity being observed. Monthly sales figures for a company which is expanding rapidly would be expected to show a steady underlying increase, possibly linear or perhaps even exponential. A company which sells greetings cards will find that the sales in some months of the year will be much higher than in others. In both cases there is an underlying deterministic pattern and some (possibly stationary) random variation on top of that. In order to predict sales figures in future months it is necessary to extrapolate the deterministic trends as well as to analyse the stationary random variation.

A further cause of non-stationarity may be that the process observed is the integrated version of a more fundamental process; in these cases, differencing the observed time series may produce a series which is more likely to be a realisation of some stationary process.

In summary, we have identified three possible causes of non-stationarity:

1. a deterministic trend (*eg* exponential or linear growth)
2. a deterministic cycle (*eg* seasonal effect)
3. the time series is integrated.

It is worth pointing out that this list is not exhaustive. For example, the first two causes are just specific cases of general deterministic behaviour. In theory, this could take many different forms, but trends and cycles are the most likely to be met in practice.

The items on the list are not mutually exclusive either. Consider a simple random walk with probability 0.6 of stepping up, and 0.4 of stepping down. This can be represented by the equation:

$$X_n = X_{n-1} + Z_n$$

where:

$$Z_n = \begin{cases} +1 & \text{with probability 0.6} \\ -1 & \text{with probability 0.4} \end{cases}$$

together with the condition that $X_0 = 0$.

This process is $I(1)$ since the first difference is stationary, but the process itself is not.

On the other hand, the process also has a increasing deterministic trend:

$$E[X_n] = E[Z_1 + Z_2 + \dots + Z_n] = E[Z_1] + E[Z_2] + \dots + E[Z_n] = 0.2n$$

We consider in the following sections how to detect and remove such causes of non-stationarity.

1.1 Detecting non-stationary series

The most useful tools in identifying non-stationarity are the simplest: a plot of the series against t , and the sample ACF.

The sample ACF is an estimate of the ACF based on sample data. It is defined in Section 2.1.

Plotting the series will highlight any obvious trends in the mean and will show up any cyclic variation which could also form evidence of non-stationarity. This should always be the first step in any practical time series analysis.



The R code below uses the `ts.plot` function.

(It also assumes we have a list of FTSE100 data values.)

```
ts.plot(log(FTSE100$Close))
points(log(FTSE100$Close), cex=.4)
```

generates Figure 14.1, which shows the time series of the logs of 300 successive closing values of FTSE100 index.

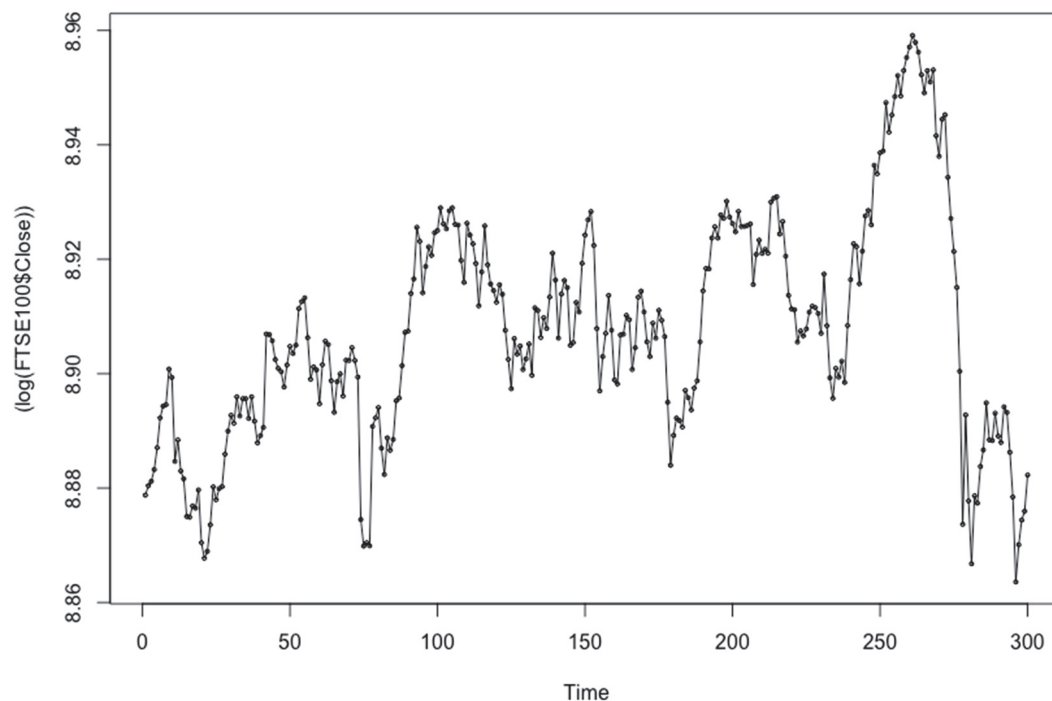


Figure 14.1: 300 successive closing values of the FTSE100 index, Jan 2017 – Mar 2018; log-transformed

The corresponding sample ACF and sample PACF are produced using:

```
par(mfrow=c(1,2))
acf(log(FTSE100$Close))
pacf(log(FTSE100$Close))
```

These are shown in Figure 14.2 below.

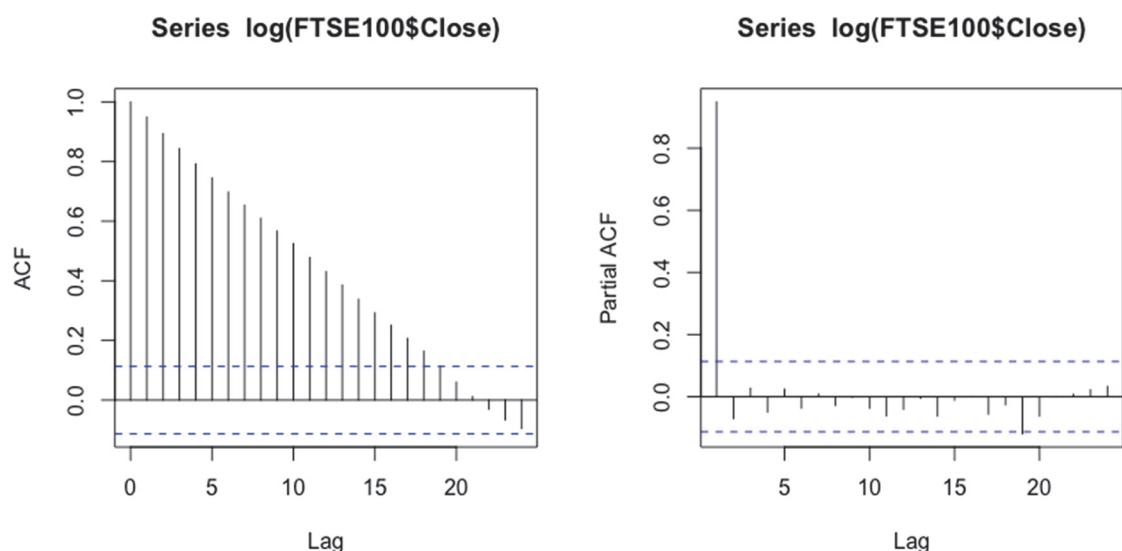


Figure 14.2: Sample ACF and sample PACF of the $\log(\text{FTSE100})$ data; dotted lines indicate cut-offs for significance if data came from some white noise process.

The sample ACF should, in the case of a stationary time series, ultimately converge towards zero exponentially fast, as for $AR(1)$ where $\rho_s = \alpha^s$.

The function $\rho_s = \alpha^s$, $s = 0, 1, 2, \dots$ has *exponential decay* if $|\alpha| < 1$ as this is a power function whose values are:

$$1, \alpha, \alpha^2, \alpha^3, \dots$$

and this sequence tends to 0 as the power s tends to ∞ .

If the sample ACF decreases slowly but steadily from a value near 1, we would conclude that the data need to be differenced before fitting the model. If the sample ACF exhibits a periodic oscillation, however, it would be reasonable to conclude that there is some underlying cause of the variation.

Figure 14.2 shows the sample ACF of a time series which is clearly non-stationary as the values decrease in some linear fashion; differencing is therefore required before fitting a stationary model.

See, for example, the change of ACF and PACF for the differenced data:

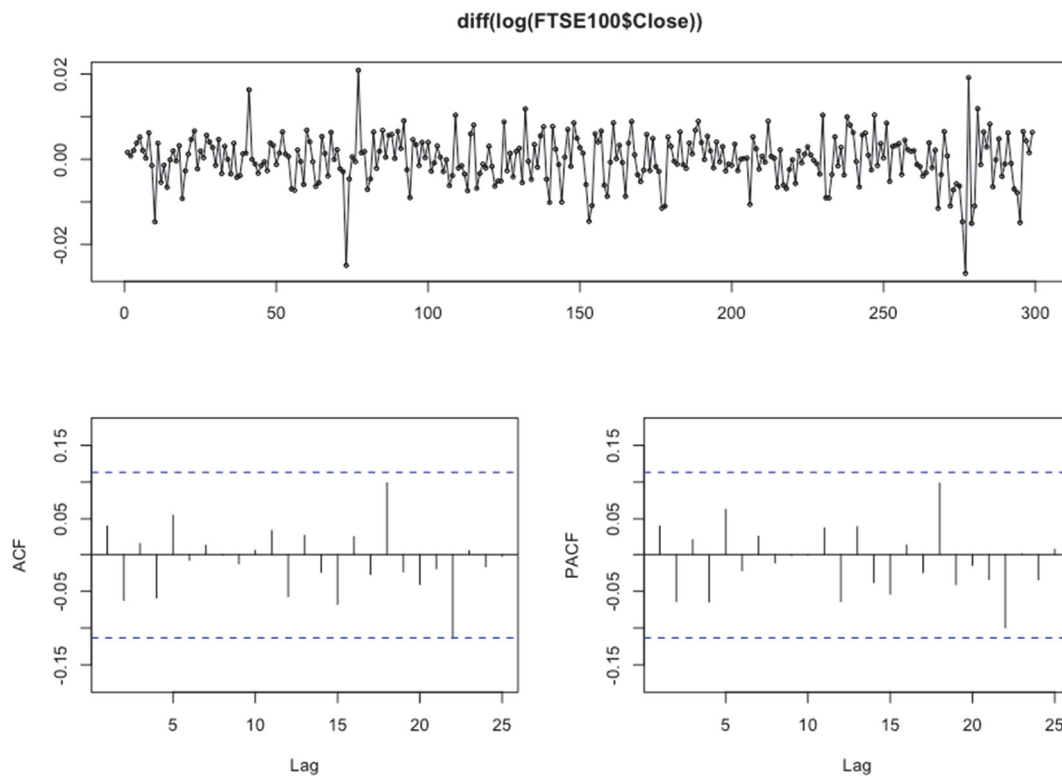


Figure 14.3: Data plot, sample ACF and sample PACF of $\nabla \ln(\text{FTSE100})$.

The ACF for a stationary $ARMA(p, q)$ process decays fairly rapidly. For example, we have seen that the ACF of a moving average process cuts off sharply, and the ACF of an $AR(1)$ process has exponential decay: $\rho(k) = \alpha^{|k|}$. In theory, this could still actually lead to a slow decay if the values of p and/or q were high, (eg the autocorrelation of an $MA(100)$ process wouldn't cut off until lag 100), but in practice the parameter values will be fairly small. If many parameters are used then the resulting model may give a good fit to the sample data, but it is unlikely to be useful for forecasting. A fairly slow decay of the sample autocorrelation function is therefore more likely to be interpreted as an indication that the time series needs to be differenced before being modelled.

If the sample autocorrelation function oscillates without decaying rapidly, as we will see in the hotel example in Section 1.4, then we might conclude that there is an underlying deterministic cycle. This would have to be removed before fitting a model to the residuals.

We now look at two methods for removing a linear trend.

1.2 Least squares trend removal

The simplest way to remove a linear trend is by ordinary least squares. This is equivalent to fitting the model:

$$x_t = a + bt + y_t$$

where a and b are constants and y is a zero-mean stationary process. The parameters a and b can be estimated by linear regression prior to fitting a stationary model to the residuals y_t .

The formulae for estimating a and b are given on page 24 of the *Tables*.

1.3 Differencing

Differencing may well be beneficial if the sample ACF decreases slowly from a value near 1, but has useful effects in other instances as well. If, for instance, $x_t = a + bt + y_t$, then:

$$\nabla x_t = b + \nabla y_t$$

so that the differencing has removed the trend in the mean.

Differencing a series d times will make an $I(d)$ series stationary. In addition however, differencing once will also remove any linear trend, as above.

On the other hand, we could remove the linear trend by using linear regression, as in Section 1.2. However, if the series is actually $I(1)$ with a trend, then least squares regression will only remove the trend. We will still be left with an $I(1)$ process that is non-stationary.

For example, consider the simple random walk discussed earlier, which has probability 0.6 of stepping up, and 0.4 of stepping down:

$$X_n = X_{n-1} + Z_n$$

where:

$$Z_n = \begin{cases} +1 & 0.6 \\ -1 & 0.4 \end{cases}$$

We have seen that this process has an increasing trend and that $E[X_n] = 0.2n$. If we let $Y_n = X_n - 0.2n$, then $E[Y_n] = 0$. So we have removed the trend. However, since:

$$Y_n = Y_{n-1} + Z_n - 0.2$$

we are still left with an $I(1)$ process that needs to be differenced in order to be stationary.

We now look at three methods for removing cycles (seasonal variation).

1.4 Seasonal differencing

Where seasonal variation is present in the data, one way of removing it is to take a seasonal difference.

Example 1

Suppose that the time series x records the monthly average temperature in London. A model of the form:

$$x_t = \mu + \theta_t + y_t \quad (14.1)$$

might be applied, where θ is a periodic function with period 12 and y is a stationary series.

Then the seasonal difference of x is defined as $(\nabla_{12}x)_t = x_t - x_{t-12}$ and we see that:

$$(\nabla_{12}x)_t = x_t - x_{t-12} = (\mu + \theta_t + y_t) - (\mu + \theta_{t-12} + y_{t-12}) = y_t - y_{t-12}$$

is a stationary process.

We can then model the seasonal difference of x as a stationary process and reconstruct the original process x itself afterwards.

We can use R to plot the time series data and to remove seasonal variation.



Figure 14.4 below is generated from the following lines in R, where functions `ts.plot`, `acf` and `pacf` are used:

```
layout(matrix(c(1,1,2,3), 2, 2, byrow = TRUE))
ts.plot(manston1$tmax, ylab="", main="Max temperatures observed at
each month (2010-2017), Manston, UK")
points(manston1$tmax, cex=0.4)
acf(manston1$tmax, main="")
pacf(manston1$tmax, main="")
```

This code assumes that there is a list of data stored as 'manston1'.

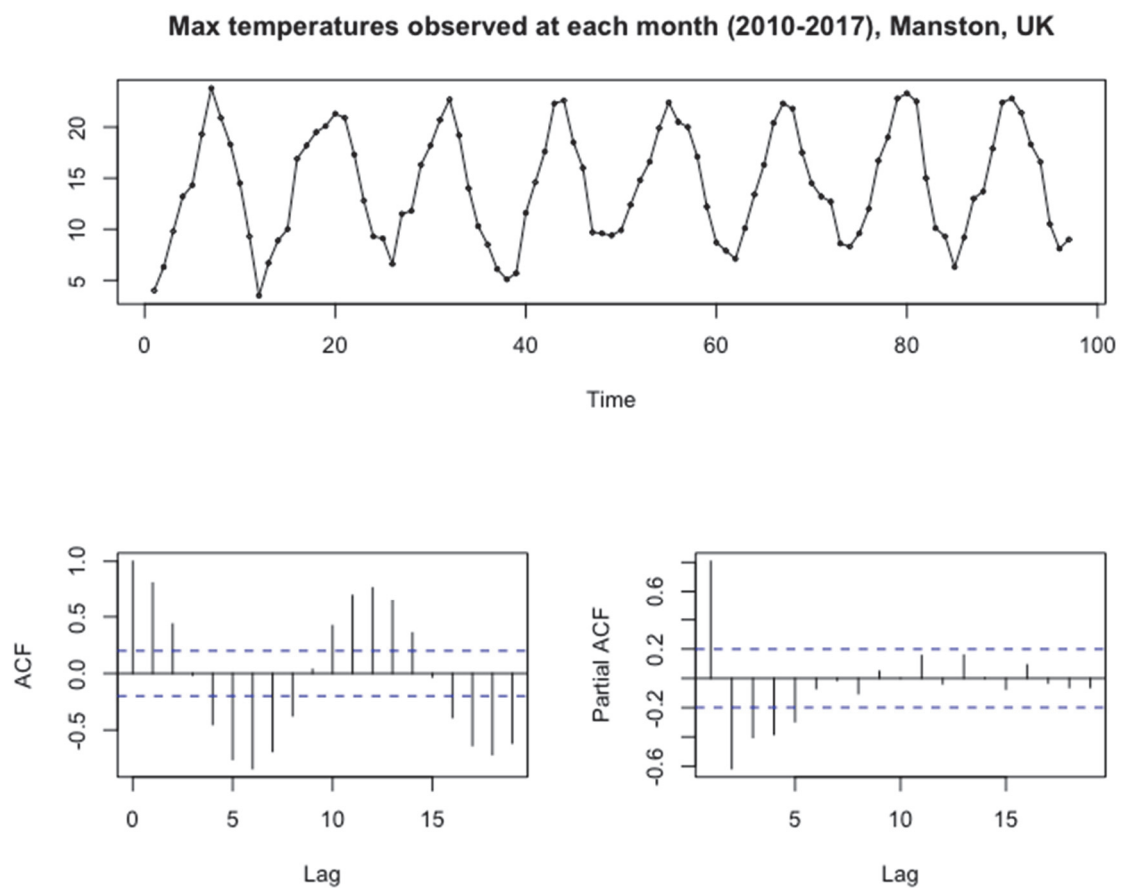


Figure 14.4: Data plot, sample ACF and PACF of temperature data.

Seasonal differencing ∇_{12} seems to have removed the seasonal behaviour of the data. See Figure 14.5 generated from:

```
layout(matrix(c(1,1,2,3), 2, 2, byrow = TRUE))
ts.plot(diff(manston1$tmax, lag=12), ylab="", main="Seasonal
differenced temperature data")
points(diff(manston1$tmax, lag=12), cex=0.4)
acf(diff(manston1$tmax, lag=12), main="")
pacf(diff(manston1$tmax, lag=12), , main="")
```

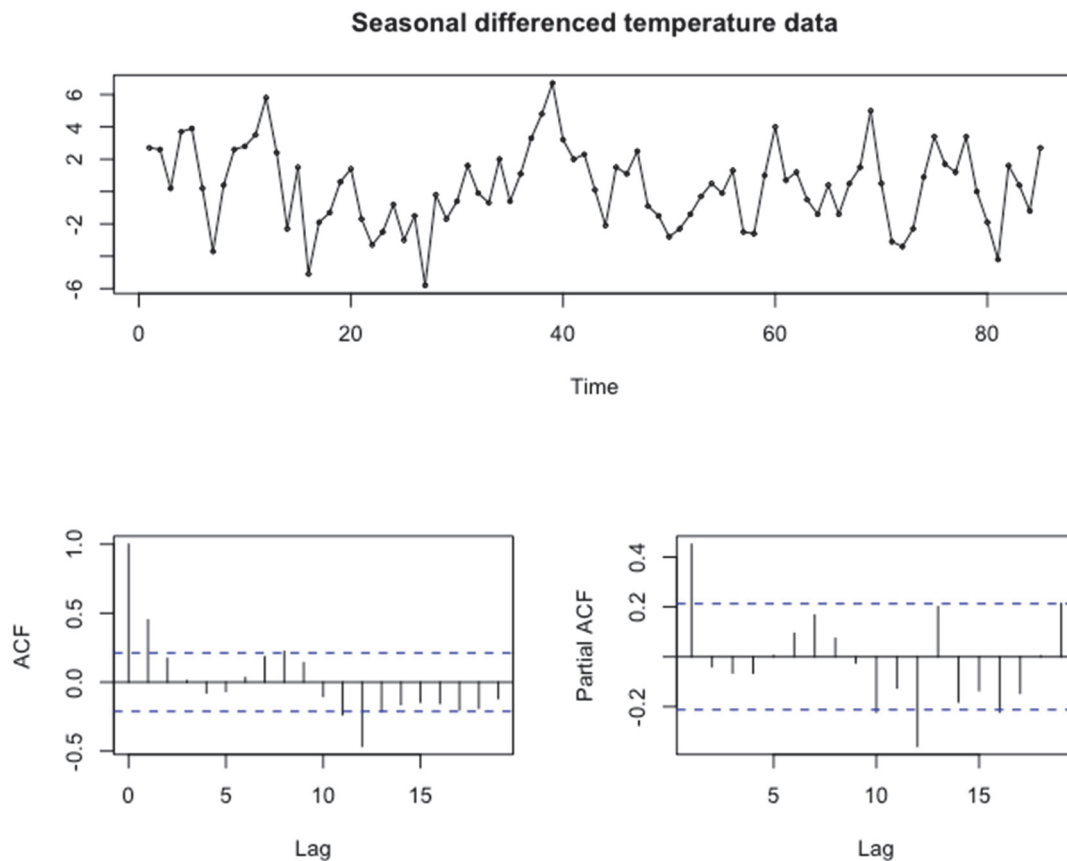


Figure 14.5: Temperature data after appropriate differencing

Example 2

The monthly inflation figures are obtained by seasonal differencing of the Retail Prices Index. If x_t is the value of the RPI in month t , the annual inflation figure reported is:

$$\frac{x_t - x_{t-12}}{x_{t-12}} \times 100\%$$

1.5 Method of moving averages

The method of moving averages makes use of a simple linear filter to eliminate the effects of periodic variation.

A linear filter is a transformation of a time series x (the input series) to create an output series y that satisfies:

$$y_t = \sum_{k=-\infty}^{\infty} a_k x_{t-k}$$

The collection of weights $\{a_k : k \in \mathbb{Z}\}$ forms a complete description of the filter. The objective of the filtering is to modify the input series to meet particular objectives, or to display specific features of the data. For example, an important problem in analysis of economic time series is detection, isolation, and removal of deterministic trends.

In practice a filter $\{a_k : k \in \mathbb{Z}\}$ normally contains only a relatively small number of non-zero components.

A very simple example of a linear filter is the difference operator $\nabla = 1 - B$. Using this filter produces:

$$y_t = (1 - B)x_t = x_t - x_{t-1}$$

So, in this case, we have $a_0 = 1$, $a_1 = -1$ and $a_k = 0$ for all other values of k .

As a second example, suppose that the input series is a white noise process e , and the filter takes the form:

$$a_0 = 1, a_1 = \beta_1, \dots, a_q = \beta_q \text{ and } a_k = 0 \text{ for all other values of } k$$

Then the output series is $MA(q)$, since we have:

$$y_t = \sum_{k=0}^q \beta_k e_{t-k}$$

Conversely, applying a filter of the form:

$$a_0 = 1, a_1 = -\alpha_1, \dots, a_p = -\alpha_p \text{ and } a_k = 0 \text{ for all other values of } k$$

to an input series x that is $AR(p)$ recovers the original white noise series:

$$y_t = x_t - \sum_{k=1}^p \alpha_k x_{t-k} = e_t$$

If x is a time series with seasonal effects with even period $d = 2h$, then we define a smoothed process y by:

$$y_t = \frac{1}{2h} \left(\frac{1}{2} x_{t-h} + x_{t-h+1} + \cdots + x_{t-1} + x_t + \cdots + x_{t+h-1} + \frac{1}{2} x_{t+h} \right)$$

This ensures that each period makes an equal contribution to y_t .

For example, with quarterly data a yearly period will have $d = 4 = 2h$, so $h = 2$ and we have:

$$y_t = \frac{1}{4} \left(\frac{1}{2} x_{t-2} + x_{t-1} + x_t + x_{t+1} + \frac{1}{2} x_{t+2} \right)$$

In this case the filter has weights:

$$a_k = \begin{cases} \frac{1}{8} & \text{for } k = -2, 2 \\ \frac{1}{4} & \text{for } k = -1, 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

This is a *centred* moving average since the average is taken symmetrically around the time t . Such a centred moving average introduces the practical problem that the average can only be calculated in retrospect, *ie* there will be a natural delay.

The same can be done with odd periods $d = 2h + 1$, but the end terms x_{t-h} and x_{t+h} do not need to be halved.

For example, with data every 4 months, a yearly period will have $d = 3 = 2h + 1$, so $h = 1$ and we have:

$$y_t = \frac{1}{3} (x_{t-1} + x_t + x_{t+1})$$

In this case the filter has weights:

$$a_k = \begin{cases} \frac{1}{3} & \text{for } k = -1, 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

As with most filtering techniques, care must be taken lest the smoothing of the data obscure the very effects which the procedure is intended to uncover.

The method of taking moving averages is one example of a series of approaches known as filtering techniques. We lose some of our knowledge about the variation in the data in exchange for (hopefully) a clearer picture of the underlying process.

1.6 Method of seasonal means

The simplest method for removing seasonal variation is to subtract from each observation the estimated mean for that period, obtained by simply averaging the corresponding observations in the sample.

Recall that the model in Equation 14.1 has the form:

$$x_t = \mu + \theta_t + y_t$$

where θ is a periodic function with period 12 and y is a stationary series. The term θ_t contains the deviation of the model at time t due to the seasonal effect. So:

$$y_t = x_t - \mu - \theta_t$$

When fitting the model in Equation 14.1 to a monthly time series x extending over 10 years from January 1990 the estimate for μ is $\bar{x}$ and the estimate for θ_{January} is:

$$\hat{\theta}_{\text{January}} = \frac{1}{10}(x_1 + x_{13} + x_{25} + \cdots + x_{109}) - \hat{\mu}$$

So, in this case, we can remove the seasonal variation by deducting the January average,

$\bar{x}_{\text{January}} = \frac{1}{10}(x_1 + x_{13} + x_{25} + \cdots + x_{109})$, from all the January values, deducting the February

average, $\bar{x}_{\text{February}} = \frac{1}{10}(x_2 + x_{14} + x_{26} + \cdots + x_{110})$, from all the February values, etc.

Alternatively, we could deduct $\hat{\theta}_{\text{January}}$ from all the January values, $\hat{\theta}_{\text{February}}$ from all the February values, etc.



In R the function `decompose` can be used to obtain both the moving average and seasonal means described in Sections 1.5 and 1.6.

```
ts.plot(manston1$tmax, ylab="", main="Max temperatures")
points(manston1$tmax, cex=0.4)
```

The time series data is plotted as in Figure 14.6 below.

```
decomp=decompose(ts(manston1$tmax, frequency = 12), type="additive")
```

The decomposition is saved as `decomp`.

The moving average can be added (in red) using the code:

```
lines(as.vector(decomp$trend), col="red")
```

The sum of seasonal and moving average trends can be added (in blue) as follows:

```
lines(as.vector(decomp$seasonal+decomp$trend), col="blue")
```

The resulting graph is shown below. A colour version is also available online in the tuition materials for the R part of CS2.

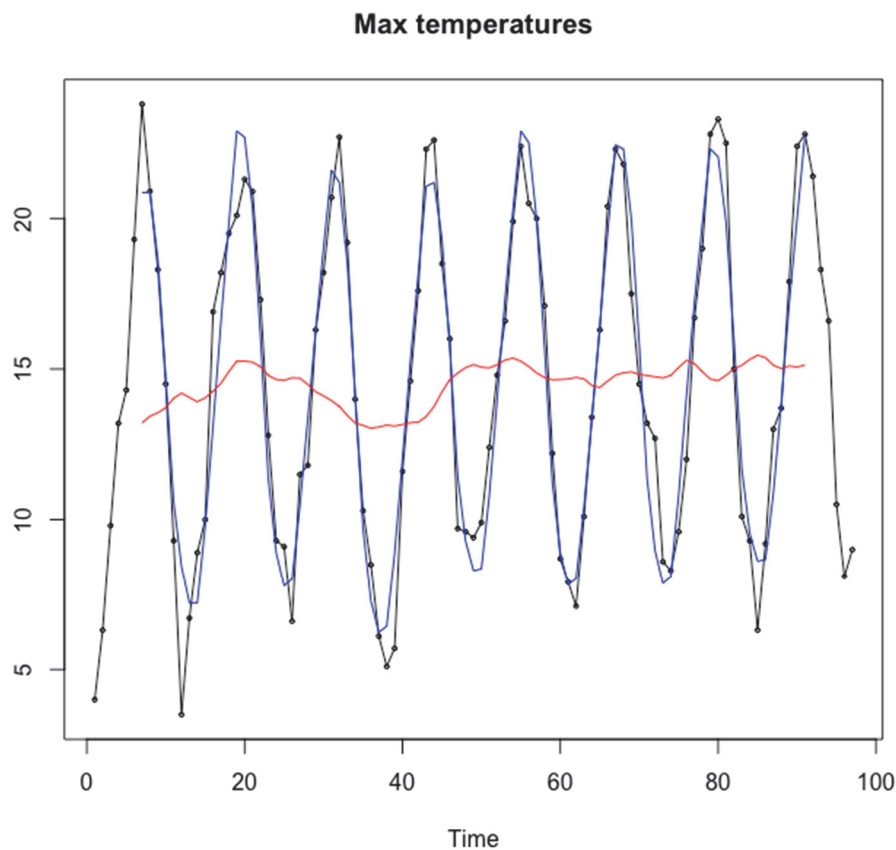


Figure 14.6: Temperature data and its decomposition into moving average (in red) and seasonal trend (in blue) added.

1.7 Transformation of the data

Diagnostic procedures such as an inspection of a plot of the residuals may suggest that even the best-fitting standard linear time series model is failing to provide an adequate fit to the data. Before attempting to use more advanced non-linear models it is often worth attempting to transform the data in some straightforward way in an attempt to find a data set on which the linear theory will work properly.

An example of a simple transformation would be $Y_t = \log X_t$, which would be used to remove an exponential growth effect.

Variance-stabilising transformations

Transformations are most commonly used when a dependence is suspected between the variance of the residuals and the size of the fitted values. If, for example, the standard deviation of $X_{t+1} - X_t$ appears to be proportional to X_t , then it would be appropriate to use the logarithmic transformation, to work on the time series $Y = \ln X$.

Transformations to increase normality

In certain applications it may be found that most residuals are small and negative, with a few large positive values to offset them. This may be taken to indicate that the distribution of the error terms is non-normal, leading to doubts as to whether the standard time series procedures, designed for normal errors, are applicable. It may be possible to find a transformation which will improve the normality of the error terms of the transformed process, but care should be taken that this does not lead to instability in the variance. A further caution when using transformed data involves the final step of turning forecasts for the transformed process into forecasts for the original process, as some transformations introduce a systematic bias.

2 Identification of $MA(q)$ and $AR(p)$ models

The treatment of this section assumes that the sequence of observations $\{x_1, x_2, \dots, x_n\}$ may be presumed to come from a stationary time series process. The problems of how to tell if the assumption of stationarity is reasonable and what to do if it is not have been treated in the previous section.

2.1 Estimation of the ACF and PACF

The autocovariance and autocorrelation functions, as seen above, play a central role in the analysis of time series. Other descriptive tools, such as the partial autocorrelation function, are derived from the ACF. Faced, then, with a sequence of observations $\{x_1, x_2, \dots, x_n\}$ and the task of finding a time series model to fit the sequence, a primary concern must be to estimate the ACF of the time series process of which the data form a realisation.

The common mean of a stationary model can be estimated using the *sample mean*:

$$\hat{\mu} = \frac{1}{n} \sum_{t=1}^n x_t$$

The autocovariance function γ_k can be estimated using the sample autocovariance function, denoted c_k or $\hat{\gamma}_k$, given by:

$$\hat{\gamma}_k = \frac{1}{n} \sum_{t=k+1}^n (x_t - \hat{\mu})(x_{t-k} - \hat{\mu})$$

from which are derived estimates r_k for the autocorrelation function ρ_k :

$$r_k = \frac{\hat{\gamma}_k}{\hat{\gamma}_0}$$

The notation $\hat{\rho}_k$ is sometimes used instead of r_k . The formulae for $\hat{\mu}$, $\hat{\gamma}_k$ and $\hat{\rho}_k$ are given on page 40 of the *Tables*.

The collection $\{r_k : k \in \mathbb{Z}\}$ is called the *sample autocorrelation function (SACF)*. Every time series analysis involves at least one plot of r_k against k . Such a plot is called a *correlogram*.

It might seem that a more natural choice for the denominator in the definition of the sample autocovariance function would be $n-k$, since there are $n-k$ terms being summed. However, the definition given is the most common one used. One theoretical reason for this is that it has a smaller mean square error at large lags, and at smaller lags the difference between the two is negligible anyway. In any case, this estimator is consistent, *ie* the bias disappears as the sample size, n , gets large.

The partial autocorrelation function ϕ_k can be estimated using the formula involving the ratio of determinants to which reference was made in [Chapter 13](#), but with the ρ_k replaced by their estimates $\hat{\rho}_k$. The resulting function $\hat{\phi}_k$, called the *sample partial autocorrelation function* (SPACF), and the plot of $\hat{\phi}_k$ against k , called the *partial correlogram*, are as important as the SACF and the correlogram in the analysis of time series.



As we have seen before, R functions `acf` and `pacf` can be used for generating these values.

For example, the following lines simulate observations from an $ARMA(1,1)$ model.

Set the seed to guarantee reproducibility. The code is:

```
set.seed(123)
```

Call the simulated data `x` :

```
x=arima.sim(n=300,model=list(ar=0.7,ma=0.5))
```

Then:

```
par(mfrow=c(1,2))
acf(x,main="Sample ACF")
pacf(x,main="Sample PACF")
```

produces the graphs below:

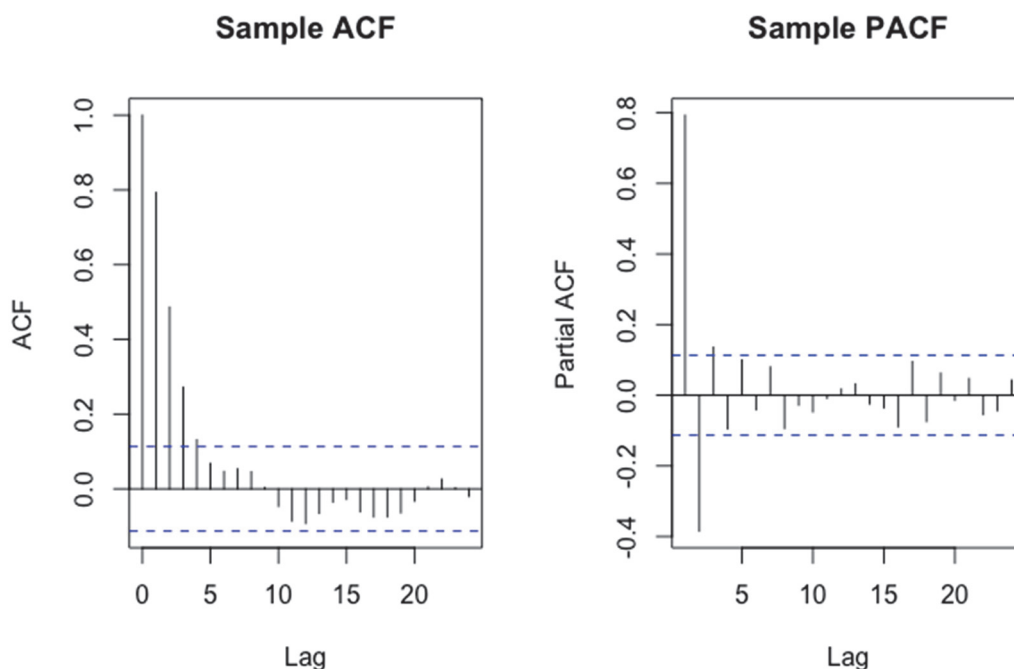


Figure 14.7: ACF and PACF of some simulated data from $ARMA(1,1)$.

2.2 Identification of white noise

A test for whether a particular sequence of observations forms a standard white noise process may seem of doubtful usefulness, but one of the techniques of residual analysis suggests that the verification of goodness of fit of any model should include a test as to whether the residuals form a white noise process. A suitable test, or portfolio of tests, is therefore a valuable asset.

We are already familiar with this idea from (normal) linear regression – we always have a look at a plot of the residuals and carry out other tests to check that the residuals do form a set of independent normal random variables.

There are many tests that could be carried out to see if a sequence of observations is a likely realisation of a white noise process. Some of these tests will be discussed later in the context of the diagnostic checking stage of the Box-Jenkins method – see [Section 3.5](#). For the moment we will concentrate on tests associated with the SACF and SPACF.

Clearly the SACF and SPACF are random, being simple functions of the observations. In particular, even if the original process was a perfectly standard white noise the SACF and SPACF would not be identically zero. The question is what scale of deviation from zero is to be expected?

An asymptotic result states that, if the original model is white noise:

$$X_t = \mu + e_t$$

then the estimators $\tilde{\rho}_k$ and $\tilde{\phi}_k$ are approximately normally distributed with mean 0, variance $1/n$ for each k .

For large samples, ie large values of n , we have the approximate distributions:

$$\tilde{\rho}_k \div N\left(0, \frac{1}{n}\right) \quad \text{and} \quad \tilde{\phi}_k \div N\left(0, \frac{1}{n}\right)$$

Values of the SACF or SPACF falling outside the range from $-2/\sqrt{n}$ to $2/\sqrt{n}$ can be taken as suggesting that the white noise model is inappropriate. This range is indicated by dashed lines in the standard output in R for ACF and PACF.

This range is an approximate 95% confidence interval based on the critical value 1.96.

But some care should be exercised: the cut-off points of $\pm 2/\sqrt{n}$ give approximate 95% limits, implying that about one value in 20 will fall outside the range even when the white noise model is correct. This means that one single value of r_k or $\hat{\phi}_k$ outside the specified range would not be regarded as significant on its own, but three such values might well be significant.

Rather than testing to see if each individual value of the SACF or SPACF lies outside a confidence interval, we can alternatively consider an overall goodness-of-fit test, much like the standard χ^2 test. In other words, we can carry out a test to see if some measure of the overall deviation over several lags lies outside some confidence interval.

A 'portmanteau' test is due to Ljung and Box, who state that, if the white noise model is correct, then:

$$n(n+2) \sum_{k=1}^m \frac{r_k^2}{n-k} \sim \chi_m^2$$

for each m .

In the result above, the notation r_k is being used to represent the estimator $\tilde{\rho}_k$ (rather than the estimate $\hat{\rho}_k$). This is a one-sided test. A large test statistic indicates that the data do *not* conform to a white noise process.



The standard commands for running these tests in R on some observations (simulated white noise here) are:

```
x <- rnorm (100)
Box.test (x, lag = 1, type = "Ljung")
```



Question

An analysis of the first 369 draws of the National Lottery gave the following SACF values:

$$r_1 = 0.100 \quad r_2 = 0.056 \quad r_3 = 0.059 \quad r_4 = 0.054 \quad r_5 = -0.005 \quad r_6 = 0.003$$

Use the portmanteau test to ascertain whether these data can be considered to be white noise.

Solution

We are testing:

H_0 : the residuals form a white noise process

against:

H_1 : the residuals do not form a white noise process

We have $m = 6$ and $n = 369$. So the observed value of the test statistic is:

$$\begin{aligned} n(n+2) \sum_{k=1}^m \frac{r_k^2}{n-k} &= 369 \times 371 \left(\frac{0.100^2}{369-1} + \frac{0.056^2}{369-2} + \frac{0.059^2}{369-3} + \frac{0.054^2}{369-4} + \frac{(-0.005)^2}{369-5} + \frac{0.003^2}{369-6} \right) \\ &= 7.298 \end{aligned}$$

We compare this with the χ_6^2 distribution. Since 7.298 is less than 12.59, the upper 5% point of χ_6^2 , there is insufficient evidence to reject H_0 at the 5% level. So we conclude that the residuals are consistent with white noise.

2.3 Identification of $MA(q)$

The distinguishing characteristic of $MA(q)$ is that $\rho_k = 0$ for all $k > q$.

A test for the appropriateness of an $MA(q)$ model, therefore, is that r_k is close to 0 for all $k > q$. If the data really do come from a $MA(q)$ model, the estimators $\tilde{\rho}_k$ for $k > q$ will be roughly normally distributed with mean 0, variance $n^{-1} \left(1 + 2 \sum_{i=1}^q \rho_i^2 \right)$.

In other words:

$$\tilde{\rho}_k \doteq N \left(0, \frac{1}{n} \left(1 + 2 \sum_{i=1}^q \rho_i^2 \right) \right)$$

This asymptotic result enables a test to be formulated.

If we assume that the data do come from a $MA(q)$ process, then an approximate 95% confidence interval for ρ_k is:

$$\left[-1.96 \sqrt{\frac{1}{n} \left(1 + 2 \sum_{i=1}^q \hat{\rho}_i^2 \right)}, +1.96 \sqrt{\frac{1}{n} \left(1 + 2 \sum_{i=1}^q \hat{\rho}_i^2 \right)} \right]$$

So no more than about 1 in 20 values of the sample autocorrelation function should lie outside the interval.

2.4 Identification of $AR(p)$

The corresponding diagnostic procedure for an autoregressive model is based on the sample partial ACF, since the PACF of an $AR(p)$ is distinctive, being equal to zero for $k > p$.

The asymptotic variance of $\tilde{\phi}_k$ is $1/n$ for each $k > p$. Again a normal approximation can be used, so that values of the SPACF outside the range $\pm 2/\sqrt{n}$ may suggest that the $AR(p)$ model is inappropriate.

So:

$$\tilde{\phi}_k \doteq N \left(0, \frac{1}{n} \right)$$

and hence an approximate 95% confidence interval for ϕ_k is:

$$\left[-1.96 \sqrt{\frac{1}{n}}, +1.96 \sqrt{\frac{1}{n}} \right]$$

As suggested by the Core Reading, the value 2 may be used as an approximation to 1.96.

3 Fitting a time series model using the Box-Jenkins methodology

In this section we consider the general class of autoregressive integrated moving average models – the $ARIMA(p,d,q)$ models. As usual we assume that historical data, comprising a time series $\{x_t : t = 1, 2, \dots, n\}$, are given.

We will also assume that deterministic trends and seasonal effects have been removed from the data, as in Section 1, although no differencing of the process is assumed – that is part of the Box-Jenkins method.

3.1 The Box-Jenkins methodology

The Box-Jenkins approach allows one to find an $ARIMA$ model which is reasonably simple and provides a sufficiently accurate description of the behaviour of the historical data.



Main steps in the Box-Jenkins approach to modelling

The main steps of the approach are:

- tentative identification of a model from the $ARIMA$ class
- estimation of parameters in the identified model
- diagnostic checks.

If the tentatively identified model passes the diagnostic tests, the model is ready to be used for forecasting. If it does not, the diagnostic tests should indicate how the model ought to be modified, and a new cycle of identification, estimation and diagnosis is performed.

The identification process is carried out in Sections 3.2 and 3.3. In Section 3.4 we look at the estimation stage, and finally, in Section 3.5 we describe some diagnostic tests that can be performed.

3.2 Differencing

An $ARIMA(p,d,q)$ model is completely identified by the choice of non-negative integer values for the parameters p , d and q . The parameter d is the number of times we have to difference the time series x to convert it to some stationary level.

In other words, once we have differenced the sample data d times, the resulting values look like a realisation of a stationary process.

We have already discussed how to detect non-stationary series in Section 1.1. Recall that there are three basic causes of non-stationarity with which we are concerned. To help identify these, we should look at plots of the data values and the SACF. The plot of the data values should highlight any obvious trends or cycles and the latter should also show up as cycles in the SACF. We are now assuming that these kind of effects have already been removed. The other cause of non-stationarity is that the time series could be the realisation of an integrated process. To remove this source of non-stationarity the sample series needs to be differenced d times.

The following principles can be used to choose the appropriate value of d :

1. A time series x can be modelled by a stationary ARMA model if the sample autocorrelation function r_k decays rapidly to zero with k . If, on the other hand, a slowly decaying positive sample autocorrelation function r_k is observed, this should be taken to indicate that the time series needs to be differenced to convert it into a likely realisation of a stationary random process.

(This is also mentioned in Section 1.1.)

2. Let $\hat{\sigma}_d^2$ denote the sample variance of the process $z^{(d)} = \nabla^d x$, ie the sample variance of the data values after they have been differenced d times. It is normally the case that $\hat{\sigma}_d^2$ first decreases with d until stationarity is achieved and then starts to increase. Therefore d can be set to the value which minimises $\hat{\sigma}_d^2$. This could be $d = 0$ if the original time series x is already stationary.



Question

The time series Q_t for the monthly RPI given on page 152 of the *Tables* has SACF:

$$r_1 = 0.977 \quad r_2 = 0.954 \quad r_3 = 0.930 \quad r_4 = 0.908 \quad \dots$$

Find an appropriate value for d in this case.

Solution

This SACF decays slowly, suggesting that it needs to be differenced. If we look at the sample variances given in the *Tables*, we find that:

$$\text{var}(Q_t) = 11.9^2$$

$$\text{var}(\nabla Q_t) = 0.6^2$$

$$\text{var}(\nabla^2 Q_t) = 0.8^2$$

This suggests that $d = 1$ would be an appropriate value.

3.3 Fitting an $ARMA(p,q)$ model

Suppose now that the appropriate value for the parameter d has been found, and the time series $\{z_{d+1}, z_{d+2}, \dots, z_n\}$ is adequately stationary. (Notice that a differenced series has d fewer observation than the original series.) We shall assume throughout this section that the sample mean of the z sequence is zero; if this is not the case, obtain a new sequence by subtracting $\hat{\mu} = \bar{z}$ from each value in the sequence. We shall also assume, for the sake of simplicity in setting down the lower and upper limits of sums, that $d = 0$.

In the framework of the Box-Jenkins approach we try to find an $ARMA(p,q)$ model which fits the data z .

If either the correlogram or the partial correlogram appears to be close to zero for sufficiently large k , an $MA(q)$ or $AR(p)$ model is indicated.

We have already seen several times that the ACF of an $MA(q)$ process cuts off (ie is equal to 0) after lag q , and similarly, the PACF of an $AR(p)$ process cuts off after lag p . We can therefore test the SACF and SPACF to see if it likely that they share one of these cut-off properties. The exact tests for this are given in Sections 2.3 and 2.4.

Otherwise we should look for an $ARMA(p,q)$ model with non-zero values of p and q .

A good indicator for possible values of p and q in an $ARMA(p,q)$ is the number of spikes in the ACF and PACF until some geometrical decay to zero is observed. Since models can be readily fitted in R, it is not hard to start with a simple model like $ARMA(1,1)$ and to work up to more complicated models if the simpler ones are deemed inadequate.

Every additional parameter improves the fit of the model by reducing the residual sum of squares. Taking this to extremes, a model with n parameters could be found to fit the data exactly.

Recall that n is the number of observed values of the time series. If we have the same number of parameters as observations, then the optimised model is a perfect fit to the data.

But this will result in some spurious model with insignificant t -values of parameter estimates and the forecasts made with such a model will be found to be practically useless. This is known as the problem of *overfitting*. The question of when to stop adding new parameters is addressed by *Akaike's information criterion* (AIC), which states that we should only consider adding an extra parameter if this results in a reduction of the residual sum of squares by a factor of at least $e^{-2/n}$, or alternatively, one can evaluate for each possible model the value of:

$$AIC(\text{model}) = \log(\hat{\sigma}^2) + 2 \times \frac{\text{number of parameters}}{n}$$

and choose as the most appropriate the one corresponding to the lowest such value.

3.4 Parameter estimation

Once the values of p and q have been identified, the problem becomes to estimate the values of parameters $\alpha_1, \alpha_2, \dots, \alpha_p$ and $\beta_1, \beta_2, \dots, \beta_q$ for the $ARMA(p,q)$ model:

$$Z_t = \alpha_1 Z_{t-1} + \alpha_2 Z_{t-2} + \dots + \alpha_p Z_{t-p} + e_t + \beta_1 e_{t-1} + \beta_2 e_{t-2} + \dots + \beta_q e_{t-q}$$

Least squares estimation suggests itself; this is equivalent to maximum likelihood estimation if the e_t may be assumed normally distributed.



Question

Consider a linear regression model of the form $Y_i = \alpha + \beta x_i + e_i$ where $e_i \sim N(0, \sigma^2)$. Show that maximum likelihood estimation and least squares estimation must result in the same parameter values.

Solution

The log-likelihood function is given by:

$$\begin{aligned}\ln L(\alpha, \beta) &= \text{const} + \sum_{i=1}^n \left\{ -\frac{1}{2} \left(\frac{y_i - \alpha - \beta x_i}{\sigma} \right)^2 \right\} \\ &= \text{const} - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \alpha - \beta x_i)^2\end{aligned}$$

It follows that maximising the log-likelihood must be equivalent to minimising the squared error:

$$\sum_{i=1}^n (y_i - \alpha - \beta x_i)^2$$

In the case of an $AR(p)$ we have:

$$e_t = z_t - \alpha_1 z_{t-1} - \dots - \alpha_p z_{t-p}$$

and the estimators $\hat{\alpha}_1, \dots, \hat{\alpha}_p$ are chosen to minimise:

$$\sum_{t=p+1}^n (z_t - \alpha_1 z_{t-1} - \dots - \alpha_p z_{t-p})^2$$

We don't need to make any distributional assumptions in order to calculate the least squares estimates of $\alpha_1, \alpha_2, \dots, \alpha_p$. However, we do need to make a distributional assumption to calculate their maximum likelihood estimates. The two methods coincide when the errors are normally distributed.

In the case of a more general $ARMA$ process we encounter the difficulty that the e_t cannot be deduced from the z_t . For example, in the case of $ARMA(1,1)$ we have:

$$e_t = z_t - \alpha_1 z_{t-1} - \beta_1 e_{t-1}$$

an equation which can be solved iteratively for e_t as long as some starting value e_0 is assumed. For an $ARMA(p,q)$ the list of starting values is $(e_0, \dots, e_{q-1})$.

The starting values need to be estimated, which is usually carried out by a recursive technique. First assume they are all equal to zero and estimate the α_i and β_j on that basis, then use standard forecasting techniques on the time-reversed process $\{z_n, \dots, z_1\}$ to obtain predicted values for $(e_0, \dots, e_{q-1})$, a method known as *backforecasting*. These new values can be used as the starting point for another application of the estimation procedure; this continues until the estimates have converged.

This would be a time consuming process to carry out by hand but such iterative procedures are easy to implement on a computer.



In Figure 14.7, the ACF and PACF plots show some significant spikes in the early lags, suggesting some presence of autoregressive and moving average.

The code:

```
fit=arima(x,order=c(1,0,1));fit
```

fits the **ARIMA(1,0,1)** to this data set, with standard output:

```
arima(x = x, order = c(1, 0, 1))
Coefficients:
      ar1      ma1  intercept
    0.6118  0.5849    0.0911
s.e.  0.0530  0.0600    0.2224
sigma^2 estimated as 0.9016:  log likelihood = -410.9,  aic = 829.8
```

where the estimated parameters **ar1** and **ma1** correspond to α and β in the **ARMA(1,1)** model. The fitted model has **AIC = 829.8**, which is the smallest value among other possible models like **AR(1)**, **MA(1)**, **ARMA(1,2)** and **ARMA(2,2)**.

An alternative method of estimation is based on method of moments estimation. There are $p+q$ parameters to be estimated. We can calculate the theoretical ACF $\{\rho_k\}$ of an ARMA (p,q) process, which will be a function of the α 's and β 's. Then the method of moments estimators are those values of α and β such that the theoretical ACF $\rho_1, \dots, \rho_{p+q}$ coincides with the observed sample ACF $r_1, \dots, r_{p+q}$. This method is easily available for **AR(p)** models since the corresponding Yule-Walker equations are linear, therefore moment estimation requires solving them with respect to the unknown parameters α_i .



Question

It is believed that a set of data values is the realisation of a **MA(1)** process $X_n = e_n + \beta e_{n-1}$ where the errors are standard normal. Given that $\hat{\gamma}_0 = 1$ and $\hat{\gamma}_1 = -0.25$, use the method of moments to estimate the parameter β , ensuring that the fitted process is invertible.

Solution

We have $r_1 = -0.25$. The theoretical values of the autocovariance at lags 0 and 1 are:

$$\gamma_0 = \text{cov}(X_t, X_t) = \text{cov}(e_t + \beta e_{t-1}, e_t + \beta e_{t-1}) = 1 + \beta^2$$

$$\gamma_1 = \text{cov}(X_t, X_{t+1}) = \text{cov}(e_t + \beta e_{t-1}, e_{t+1} + \beta e_t) = \beta$$

So $\rho_1 = \frac{\beta}{1 + \beta^2}$. Equating this with the corresponding sample correlation at lag 1 gives:

$$\rho_1 = \frac{\beta}{1 + \beta^2} = \frac{-0.25}{1} \Rightarrow \beta^2 + 4\beta + 1 = 0$$

Solving this quadratic we get $\beta = -0.268$ or -3.732 .

Setting $\beta = -0.268$ gives an invertible process; setting $\beta = -3.732$ does not. So we take -0.268 to be our estimated value of β .

The final parameter of the model is σ^2 , the variance of the e_t , which may be estimated using:

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{t=p+1}^n \hat{e}_t^2 = \frac{1}{n} \sum_{t=p+1}^n (z_t - \hat{\alpha}_1 z_{t-1} - \dots - \hat{\alpha}_p z_{t-p} - \hat{\beta}_1 \hat{e}_{t-1} - \dots - \hat{\beta}_q \hat{e}_{t-q})^2$$

where $\hat{e}_t$ denotes the residual at time t .

The residuals are the realised values of the white noise terms.

The parameter σ^2 needs to be estimated last as the formula above involves the estimated values of the α 's and β 's. σ^2 has not been used to estimate the other parameters, whichever method is used, so the overall estimation procedure is well-defined. However, if we use the method of moments to estimate the α 's and β 's, we would have to estimate the errors separately (as we would not have estimated these as part of the procedure for the estimation of the α 's and β 's).

If the number of observations n of the time series is sufficiently large there will be little difference between the least squares estimates and the method of moments estimates of the parameters.

3.5 Diagnostic checking

After the tentative identification of an $ARIMA(p, d, q)$ model and calculation of the estimates $\hat{\mu}, \hat{\sigma}, \hat{\alpha}_1, \dots, \hat{\alpha}_p, \hat{\beta}_1, \dots, \hat{\beta}_q$ we have to perform diagnostic checking. The principle of this is that, if the $ARMA(p, q)$ model is a good approximation to the underlying time series process, then the residuals $\hat{e}_t$ will form a good approximation to a white noise process.

One test that we could perform on the residuals is the Ljung and Box portmanteau test, which is covered in Section 2.2. There are other tests, however, some of which are outlined below.

The following checks are frequently used.

Inspection of the graph of the residuals

The visual inspection of the graph of the residuals against t or the graph of $\hat{e}_t$ against z_t can help to highlight a poorly fitting model. If any pattern is evident, whether in the average level of the residuals or in the magnitude of the fluctuations about 0, this should be taken to mean that the model is inadequate.

The question is really whether the errors are a likely realisation of a set of independent normal random variables. So a 'pattern' is anything that may suggest non-independence.

We could be slightly more quantitative with this. Assuming the errors are independent normal random variables there should be a certain proportion of the sample errors within any given range. For example, if the errors were thought to be standard normal, then we would expect 34% of the sample values to lie between 0 and 1.

Inspection of the sample autocorrelation functions of the residuals

The behaviour of the sample ACF and sample PACF of a white noise sequence have already been described.

This was done in Section 2.2.

If the SACF or SPACF of the sequence of residuals has too many values outside the range $\pm 2/\sqrt{N}$, we conclude that the fitted model does not have enough parameters and a new model with additional parameters should be fitted. The Ljung-Box chi-squared statistic may also be used for this purpose, but the degrees of freedom of the test statistics needs to be reduced by the number of parameters $p + q$ of the ARMA model.

Here the Core Reading is using N rather than n to denote the number of recorded values of the time series.

Counting turning points

If $y_1, y_2, \dots, y_N$ is a sequence of numbers, then we say that the sequence has a *turning point* at time k if either $y_{k-1} < y_k$ and $y_k > y_{k+1}$, or $y_{k-1} > y_k$ and $y_k < y_{k+1}$.

If $Y_1, Y_2, \dots, Y_N$ is a sequence of independent random variables with continuous distribution, then the probability of a turning point at time k is $\frac{2}{3}$, the expected number of turning points is $\frac{2}{3}(N-2)$, and the variance is $\frac{16N-29}{90}$.

This is a result for a sequence of *independent* random variables. It is therefore usually applied to the *residuals* of the time series, not to the original time series itself, which will not be independent. The procedure is therefore to calculate the residuals, and then to apply the turning point test to them to see if the residuals have a reasonable number of turning points.

A proof of this result is beyond the scope of Subject CS2. The formulae for the mean and variance of the number of turning points are both on page 42 of the *Tables*.

Therefore the number of turning points in a realisation of $Y_1, Y_2, \dots, Y_N$ should be within the 95% confidence interval:

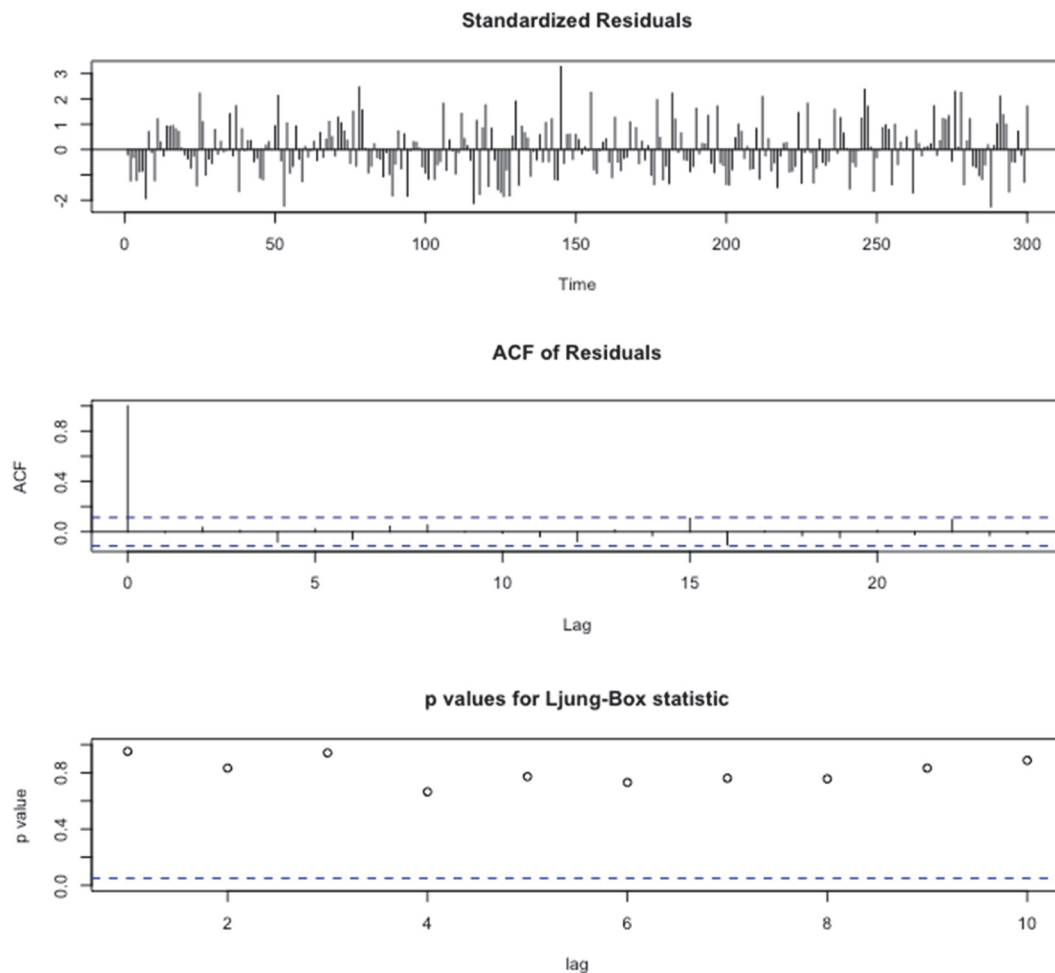
$$\left[\frac{2}{3}(N-2) - 1.96\sqrt{\frac{16N-29}{90}}, \frac{2}{3}(N-2) + 1.96\sqrt{\frac{16N-29}{90}} \right]$$



The command:

```
tsdiag(fit)
```

generates a graphical summary of the diagnostic checks of the residuals.



where the last plot shows a sequence of p -values of the Ljung-Box test, high values observed suggesting good fit, ie residuals close to white noise.

Figure 14.8: Diagnostic checks of residuals

4 Forecasting

4.1 Box-Jenkins approach to forecasting stationary time series

Using the Box-Jenkins approach, forecasting is relatively straightforward. Having fitted an ARMA model to the data $\{x_1, \dots, x_n\}$ we have the equation:

$$X_{n+k} = \mu + \alpha_1(X_{n+k-1} - \mu) + \dots + \alpha_p(X_{n+k-p} - \mu) + e_{n+k} + \beta_1 e_{n+k-1} + \dots + \beta_q e_{n+k-q}$$

Recall that at the start of Section 3.3, it was assumed that the data set to which we were fitting the ARMA model had zero mean. In the expression above the mean μ has now been put back in.



Forecasting future values of an ARMA process

The forecast value of X_{n+k} given all observations up until time n , known as the k -step ahead forecast and denoted $\hat{x}_n(k)$, is obtained from this equation by:

- replacing all (unknown) parameters by their estimated values;
- replacing the random variables $X_1, \dots, X_n$ by their observed values $x_1, \dots, x_n$;
- replacing the random variables $X_{n+1}, \dots, X_{n+k-1}$ by their forecast values $\hat{x}_n(1), \dots, \hat{x}_n(k-1)$;
- replacing the innovations $e_1, \dots, e_n$ by the residuals $\hat{e}_1, \dots, \hat{e}_n$;
- replacing the random variables $e_{n+1}, \dots, e_{n+k-1}$ by their expectations, 0.

For example, the one-step ahead and two-step ahead forecasts for an $AR(2)$ are given by:

$$\hat{x}_n(1) = \hat{\mu} + \hat{\alpha}_1(x_n - \hat{\mu}) + \hat{\alpha}_2(x_{n-1} - \hat{\mu})$$

$$\hat{x}_n(2) = \hat{\mu} + \hat{\alpha}_1(\hat{x}_n(1) - \hat{\mu}) + \hat{\alpha}_2(x_n - \hat{\mu})$$



Question

Write down an expression for the two-step ahead forecast of a general $ARMA(2,2)$ process.

Solution

The two-step ahead forecast is:

$$\hat{x}_n(2) = \hat{\mu} + \hat{\alpha}_1(\hat{x}_n(1) - \hat{\mu}) + \hat{\alpha}_2(x_n - \hat{\mu}) + \hat{\beta}_2 \hat{e}_n$$

Thus the k -step ahead forecast is essentially the conditional expectation of the future value of the process given all the information currently available at time n .

A point estimate of X_{n+k} is less useful than a confidence interval, for which an estimate of the variance is required. A comparison of X_{n+1} with $\hat{x}_n(1)$ shows that the difference between them arises from numerous sources, including e_{n+1} , differences between true values of parameters and their estimates, and differences between true values of the e_t and the residuals $\hat{e}_t$ which are used to estimate them. Calculation of the prediction variance in any given case is complicated and is best left to a computer. In general, though, it is possible to state that the variance of the k -step ahead estimator is relatively small for small values of k and converges, for large k , to γ_0 , the variance of the stationary process X .



Question

Explain why the k -step ahead variance converges toward the variance of the process.

Solution

The process is stationary. So starting from time n and projecting into the future, the process eventually settles down into the equilibrium distribution.

4.2 Forecasting ARIMA processes

If X is an $ARIMA(p, d, q)$ process, then $Z = \nabla^d X$ is $ARMA(p, q)$, so the techniques of Section 4.1 can be used to produce forecasts and confidence intervals for future values of Z . By reversing the differencing procedure these can be translated into forecasts of future values of X .

For example, suppose that X is $ARIMA(0, 1, 1)$.

Then $Z_n = \nabla X_n = X_n - X_{n-1}$ is $ARMA(0, 1)$, and $X_n = X_{n-1} + Z_n$.

Hence $X_{n+1} = X_n + Z_{n+1}$, and $\hat{x}_n(1) = x_n + \hat{z}_n(1)$.



Question

Give an expression for the two-step ahead forecast of an $ARIMA(1, 2, 1)$ process.

Solution

If we define:

$$Z_n = \nabla^2 X_n = (1 - B)^2 X_n = (1 - 2B + B^2)X_n = X_n - 2X_{n-1} + X_{n-2}$$

This implies that:

$$X_n = Z_n + 2X_{n-1} - X_{n-2}$$

Hence:

$$X_{n+2} = 2X_{n+1} - X_n + Z_{n+2}$$

So the two-step ahead forecast is:

$$\hat{x}_n(2) = 2\hat{x}_n(1) - x_n + \hat{z}_n(2)$$

Since Z is $ARMA(1,1)$, it has a defining equation of the form:

$$Z_n = \mu + \alpha(Z_{n-1} - \mu) + e_n + \beta e_{n-1}$$

So:

$$Z_{n+2} = \mu + \alpha(Z_{n+1} - \mu) + e_{n+2} + \beta e_{n+1}$$

and:

$$\hat{z}_n(2) = \hat{\mu} + \hat{\alpha}(\hat{z}_n(1) - \hat{\mu})$$

Hence the two-step ahead forecast can be expressed as:

$$\hat{x}_n(2) = 2\hat{x}_n(1) - x_n + \hat{\mu} + \hat{\alpha}(\hat{z}_n(1) - \hat{\mu})$$

An $ARIMA(p,d,q)$ process with $d > 0$ is not stationary and therefore has no stationary variance. It should come as no surprise, then, that the prediction variance for the k -step ahead forecast increases to infinity as k increases. This is easily seen in the case of the random walk process.



For predicting three steps ahead:

```
predict(fit, n.ahead=3)
```

4.3 Exponential smoothing

The Box-Jenkins methodology is demanding, requiring a skilled operator to produce reliable results. There are many instances in which a company needs no more than a simple forecast of some future value without having to employ a trained statistician to provide it. A much simpler forecasting technique, introduced by Holt in 1958, uses a weighted combination of past values to predict future observations.



One-step ahead forecast using exponential smoothing

$$\hat{x}_n(1) = \alpha(x_n + (1-\alpha)x_{n-1} + (1-\alpha)^2x_{n-2} + \dots)$$

The weights used here are $\alpha, \alpha(1-\alpha), \alpha(1-\alpha)^2, \dots$.

Here α is a single parameter, either chosen by the user or estimated by least squares from past data. Typically a value in the range 0.2 to 0.3 is used.

A value of α between 0 and 1 will give a weighted average of historic values with less emphasis on values that are further back in time.

The geometrically decreasing weights give rise to the name *exponential smoothing*.

Since the weights sum to 1, the exponential smoothing filter is a weighted average of historic values, with the weights decreasing geometrically as we go further back in time.



Question

Show that the weights sum to 1 when $0 < \alpha < 1$.

Solution

The weights form a geometric progression with first term α and common ratio $1 - \alpha$. Using the formula for the sum to infinity of a geometric progression, we see that:

$$\alpha + \alpha(1 - \alpha) + \alpha(1 - \alpha)^2 + \dots = \frac{\alpha}{1 - (1 - \alpha)} = 1$$

The method lends itself easily to regular updating. It is easy to see that:

$$\hat{x}_n(1) = (1 - \alpha)\hat{x}_{n-1}(1) + \alpha x_n = \hat{x}_{n-1}(1) + \alpha[x_n - \hat{x}_{n-1}(1)]$$

so that the current forecast is obtained by taking the previous forecast and compensating for the error observed when the actual figure became available.

This technique works for stationary series, but clearly cannot be applied to series exhibiting a trend or seasonal variation. There are more sophisticated versions of exponential smoothing which are able to cope with trends or seasonal variation, and are even well equipped to handle slowly varying trends or multiplicative, rather than additive, seasonal variation.

5 Multivariate time series models

5.1 Vector autoregressions

An m -dimensional multivariate time series $\{\underline{x}_1, \dots, \underline{x}_n\}$ is a sequence of m -dimensional vectors. Each vector $\underline{x}_t$ is a set of observations of the values of m variables of interest at time t . A multivariate time series is modelled by a sequence of random vectors $\{\underline{X}_1, \underline{X}_2, \dots\}$. The components of $\underline{X}_t$ will be denoted $X_t^{(1)}, \dots, X_t^{(m)}$.

The second-order properties of a sequence of random vectors are summarised by:

- the *vectors of expected values* $\underline{\mu}_t = E[\underline{X}_t]$, and
- the *covariance matrices* for all pairs of random vectors, $\text{cov}(\underline{X}_t, \underline{X}_{t+k})$.

The definition of stationarity is the same in the multidimensional case as it is for univariate time series: the vector process is (weakly) stationary if both $E[\underline{X}_t]$ and $\text{cov}(\underline{X}_t, \underline{X}_{t+k})$ are independent of t . In the stationary case, the notation $\underline{\mu}$ will be used to represent the common mean vector, Σ_k the covariance matrix $\text{cov}(\underline{X}_t, \underline{X}_{t+k})$.

The diagonal elements of the covariance matrix Σ_k are clearly the autocovariances at lag k of the individual components of the random vector $\underline{X}_t$. The off-diagonal elements $\Sigma_k(i, j)$ are called the lag k *cross-covariances* of $X_t^{(i)}$ with $X_{t+k}^{(j)}$, $\text{cov}(X_t^{(i)}, X_{t+k}^{(j)})$.

Example 1

A multivariate white noise process is the simplest example of a multivariate random process. Suppose $\underline{e}_1, \underline{e}_2, \dots$ is a sequence of independent zero-mean random vectors, each having the same covariance matrix Σ .

Σ need not be a diagonal matrix, though it must be symmetrical. In other words, the components of the innovations vector need not be independent of one another. This is a multivariate analogue of the zero-mean white noise.

A *vector autoregressive process* of order p , denoted $VAR(p)$, is a sequence of m -component random vectors $\{\underline{X}_1, \underline{X}_2, \dots\}$ satisfying:

$$\underline{X}_t = \underline{\mu} + \sum_{j=1}^p A_j (\underline{X}_{t-j} - \underline{\mu}) + \underline{e}_t \quad (14.2)$$

where $\underline{e}$ is an m -dimensional white noise process and the A_j are $m \times m$ matrices.

So a $VAR(1)$ process has the following structure:

$$\underline{X}_t - \underline{\mu} = A(\underline{X}_{t-1} - \underline{\mu}) + \underline{e}_t$$

Example 2

We might believe that interest rates, i_t , and tendency to invest, I_t , are related to one another by the equations:

$$\begin{cases} i_t - \mu_i &= \alpha_{11}(i_{t-1} - \mu_i) + e_t^{(i)} \\ I_t - \mu_I &= \alpha_{21}(i_{t-1} - \mu_i) + \alpha_{22}(I_{t-1} - \mu_I) + e_t^{(I)} \end{cases} \quad (14.3)$$

where $e_t^{(i)}$ and $e_t^{(I)}$ are zero-mean (univariate) white noises. They may have different variances and are not necessarily uncorrelated; that is, we do not require $\text{cov}(e_t^{(i)}, e_t^{(I)}) = 0$, although we do require $\text{cov}(e_t^{(i)}, e_s^{(I)}) = 0$ for $s \neq t$.

This model can be expressed as a 2-dimensional VAR(1) process:

$$\begin{pmatrix} i_t - \mu_i \\ I_t - \mu_I \end{pmatrix} = \begin{pmatrix} \alpha_{11} & 0 \\ \alpha_{21} & \alpha_{22} \end{pmatrix} \begin{pmatrix} i_{t-1} - \mu_i \\ I_{t-1} - \mu_I \end{pmatrix} + \begin{pmatrix} e_t^{(i)} \\ e_t^{(I)} \end{pmatrix}$$

The theory and analysis of a VAR(1) closely parallels that of a univariate AR(1). Iterating from Equation (14.2) in the case $p = 1$, it is clear that:

$$\underline{X}_t = \underline{\mu} + \sum_{j=0}^{t-1} A^j \underline{e}_{t-j} + A^t (\underline{X}_0 - \underline{\mu})$$

In order that $\underline{X}$ should represent a stationary time series, the powers of A should converge to zero in some sense. The appropriate requirement is that all eigenvalues of the matrix A should be less than 1 in absolute value.



Condition for stationarity of a VAR(1) process

A process of the form:

$$\underline{X}_t - \underline{\mu} = A(\underline{X}_{t-1} - \underline{\mu}) + \underline{e}_t$$

is stationary if all the eigenvalues of the matrix A are strictly less than 1 in magnitude.

The eigenvalues of matrix A are the values λ such that $\det(A - \lambda I) = 0$ where I is the identity matrix.

Eg for a 2-dimensional time series this equation reduces to:

$$(\alpha_{11} - \lambda)(\alpha_{22} - \lambda) - \alpha_{12}\alpha_{21} = 0$$

where $A[i, j] = \alpha_{ij}$ ($i = 1, 2, j = 1, 2$).

This is because we require:

$$\det(A - \lambda I) = \det\left(\begin{pmatrix} \alpha_{11} & \alpha_{12} \\ \alpha_{21} & \alpha_{22} \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right) = \det\begin{pmatrix} \alpha_{11} - \lambda & \alpha_{12} \\ \alpha_{21} & \alpha_{22} - \lambda \end{pmatrix} = 0$$

The appendix to this chapter briefly revises the definition of an eigenvalue.



Question

Determine whether the following multivariate time series is stationary:

$$\begin{pmatrix} X_t \\ Y_t \end{pmatrix} = \begin{pmatrix} 0.3 & 0.5 \\ 0.2 & 0.2 \end{pmatrix} \begin{pmatrix} X_{t-1} \\ Y_{t-1} \end{pmatrix} + \begin{pmatrix} e_t^X \\ e_t^Y \end{pmatrix}$$

Solution

To check stationarity, we need to calculate the eigenvalues of the matrix $\begin{pmatrix} 0.3 & 0.5 \\ 0.2 & 0.2 \end{pmatrix}$, ie we need to determine the values of λ for which:

$$\det\begin{pmatrix} 0.3 - \lambda & 0.5 \\ 0.2 & 0.2 - \lambda \end{pmatrix} = (0.3 - \lambda)(0.2 - \lambda) - 0.1 = \lambda^2 - 0.5\lambda - 0.04 = 0$$

The solutions of this equation are $\lambda = 0.57$ and $\lambda = -0.07$. Since both eigenvalues are strictly less than 1 in magnitude, the process is stationary.

The matrix equation given in the previous question can be written as the pair of equations:

$$X_t = 0.3X_{t-1} + 0.5Y_{t-1} + e_t^X$$

$$Y_t = 0.2X_{t-1} + 0.2Y_{t-1} + e_t^Y$$

Rearranging the first equation we get:

$$Y_{t-1} = 2(X_t - 0.3X_{t-1} - e_t^X)$$

We can substitute this in the second equation for Y_{t-1} and the similar expression for Y_t that it implies. After tidying up we have:

$$X_{t+1} = 0.5X_t + 0.04X_{t-1} + e_{t+1}^X - 0.2e_t^X + 0.5e_t^Y$$

We can check for stationarity by looking at the roots of the characteristic equation of the autoregressive part:

$$1 - 0.5\lambda - 0.04\lambda^2 = 0$$

This has roots -14.25 and 1.75 . Since both roots are strictly greater than 1 in magnitude, the process X is stationary in its own right. Note that these roots are the reciprocals of the eigenvalues calculated in the previous question.

Similar, though more complicated, requirements can be set out under which a more general $VAR(p)$ process is stationary.

Fitting a vector autoregression is very similar to the process of fitting a univariate autoregression. Parameter estimation can be carried out by least squares or by method of moments. Some elements of the univariate theory, such as the use of Akaike's Information Criterion, do not translate unchanged into a multivariate setting, but other topics carry across relatively easily.

Example

The following simple dynamic Keynesian model provides an example of a multivariate autoregressive process.

Keynesian models are studied in economics. Here we are not really interested in the economic theory that leads to this model. The important point is to understand the vector time series equations.

Denote by Y_t the national income over a certain period of time, and denote by C_t and I_t the total consumption and investment over the same period. It is assumed that the consumption, C_t , depends on the income over the previous period:

$$C_t = \alpha Y_{t-1} + e_t^{(1)}$$

where $e_t^{(1)}$ is a zero-mean white noise. The investment, I_t , is determined by the 'accelerator' mechanism:

$$I_t = \beta (C_{t-1} - C_{t-2}) + e_t^{(2)}$$

where $e_t^{(2)}$ is another zero-mean white noise. Finally, any part of the national income is either consumed or invested; therefore:

$$Y_t = C_t + I_t$$

Eliminating the national income we arrive at the following two-dimensional $VAR(2)$ process:

$$C_t = \alpha C_{t-1} + \alpha I_{t-1} + e_t^{(1)}$$

$$I_t = \beta (C_{t-1} - C_{t-2}) + e_t^{(2)}$$

Using matrix notation we can rewrite the above equation as:

$$\begin{pmatrix} C_t \\ I_t \end{pmatrix} = \begin{pmatrix} \alpha & \alpha \\ \beta & 0 \end{pmatrix} \begin{pmatrix} C_{t-1} \\ I_{t-1} \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ -\beta & 0 \end{pmatrix} \begin{pmatrix} C_{t-2} \\ I_{t-2} \end{pmatrix} + \begin{pmatrix} e_t^{(1)} \\ e_t^{(2)} \end{pmatrix}$$

5.2 Cointegrated time series

Recall that a time series process X is called integrated of order d , abbreviated as $I(d)$, if the process $Y = \nabla^d X$ is stationary.



Cointegrated series

Two time series processes X and Y are called *cointegrated* if:

- (i) X and Y are $I(1)$ random processes,
- (ii) there exists a non-zero vector (α, β) such that $\alpha X + \beta Y$ is stationary.

The vector (α, β) is called a *cointegrating vector*.

In other words, two processes are themselves non-stationary (technically $I(1)$), but their movements are correlated in such a way that a certain weighted average of the two processes is stationary.

There are a number of circumstances when it is reasonable to expect that two processes may be cointegrated:

- if one of the processes is driving the other
- if both are being driven by the same underlying process.

We now consider an example of cointegrated processes. Understanding the underlying economic theory is not so important here. However, it is important to understand why the processes are cointegrated.

Example

The following simple model of evolution of the USDollar/GBPound exchange rate X_t provides an example of a cointegrated model. It is assumed that the exchange rate fluctuates around the purchasing power P_t/Q_t , where P_t and Q_t are the consumer price indices for US and UK, respectively.

This is described by the following model:

$$\ln X_t = \ln \frac{P_t}{Q_t} + Y_t$$

$$Y_t = \mu + \alpha(Y_{t-1} - \mu) + e_t + \beta e_{t-1}$$

where e is a zero-mean white noise process.

The evolution of $\ln P$ and $\ln Q$ is described by $ARIMA(1,1,0)$ models:

$$(1-B)\ln P_t = \mu_1 + \alpha_1[(1-B)\ln P_{t-1} - \mu_1] + e_t^{(1)}$$

$$(1-B)\ln Q_t = \mu_2 + \alpha_2[(1-B)\ln Q_{t-1} - \mu_2] + e_t^{(2)}$$

where $e^{(1)}$ and $e^{(2)}$ are zero-mean white noises, possibly correlated.

$\ln P$ and $\ln Q$ are both $ARIMA(1,1,0)$ processes. The logarithm of the exchange rate is also non-stationary. However:

$$\ln X - \ln P + \ln Q$$

is the $ARMA(1,1)$ random process Y and, therefore, is a stationary random process.

Here we are assuming that $|\alpha| < 1$.

It follows that the sequence of random vectors:

$$\{(\ln X_t, \ln P_t, \ln Q_t) : t = 1, 2, \dots\}$$

is described by a cointegrated model with the cointegrating vector $(1, -1, 1)$.



Question

Two time series X and Y are defined by the equations:

$$X_t = 0.65X_{t-1} + 0.35Y_{t-1} + e_t^X$$

$$Y_t = 0.35X_{t-1} + 0.65Y_{t-1} + e_t^Y$$

where e^X and e^Y are independent white noise processes.

Show that X and Y are cointegrated, with cointegrating vector $(1, -1)$.

Solution

We begin by showing that the processes X and Y are $I(1)$.

From the first equation we have:

$$Y_{t-1} = \frac{1}{0.35} (X_t - 0.65X_{t-1} - e_t^X)$$

Using this in the second equation gives:

$$\frac{1}{0.35} (X_{t+1} - 0.65X_t - e_{t+1}^X) = 0.35X_{t-1} + 0.65 \frac{1}{0.35} (X_t - 0.65X_{t-1} - e_t^X) + e_t^Y$$

Tidying up we have:

$$X_{t+1} = 1.3X_t - 0.3X_{t-1} + e_{t+1}^X - 0.65e_t^X + 0.35e_t^Y$$

or equivalently:

$$X_t = 1.3X_{t-1} - 0.3X_{t-2} + e_t^X - 0.65e_{t-1}^X + 0.35e_{t-1}^Y$$

The characteristic polynomial of the AR part of this equation is:

$$1 - 1.3\lambda + 0.3\lambda^2$$

The roots of this equation are $\frac{10}{3}$ and 1. So X is not stationary. Differencing once will eliminate the root of 1. Since the only other root is strictly greater than 1 in magnitude, ∇X is a stationary process. Hence X is $I(1)$.

The process Y has a similar structure to that of X , so Y is also $I(1)$.

We now need to show that $X - Y$ is stationary.

We have equations:

$$X_t = 0.65X_{t-1} + 0.35Y_{t-1} + e_t^X$$

$$Y_t = 0.35X_{t-1} + 0.65Y_{t-1} + e_t^Y$$

Subtracting the second equation from the first we see that:

$$X_t - Y_t = 0.3X_{t-1} - 0.3Y_{t-1} + e_t^X - e_t^Y = 0.3(X_{t-1} - Y_{t-1}) + e_t^X - e_t^Y$$

Setting $W_t = X_t - Y_t$, this is:

$$W_t = 0.3W_{t-1} + (e_t^X - e_t^Y)$$

The process W is a stationary $AR(1)$ process since the root of its characteristic equation is $\frac{10}{3}$, and this is greater than 1 in magnitude. (As always, the white noise terms don't affect the stationarity.)

6 Some special non-stationary and non-linear time series models

Although the ARIMA class of processes is the most important for us, there are many other types of model. This section briefly discusses a few of them.

6.1 Bilinear models

The general class of bilinear models can be exemplified by its simplest representative, the random process X defined by the relation:

$$X_n - \alpha(X_{n-1} - \mu) = \mu + e_n + \beta e_{n-1} + b(X_{n-1} - \mu)e_{n-1}$$

Considered only as a function of X , this relation is linear; it is also linear when considered as a function of e only. This is why it is called 'bilinear'.

The main qualitative difference between the bilinear model and models from the ARMA class is that many bilinear models exhibit 'bursty' behaviour: when the process is far from its mean it tends to exhibit larger fluctuations. The difference between this model and an ARMA(1,1) process may be seen to lie in the last term on the right-hand side: when X_{n-1} is far from μ and e_{n-1} is far from 0 – events which are far from being independent – the final term assumes a much greater significance.

6.2 Threshold autoregressive models

A simple representative of the class of threshold autoregressive models is the random process X defined by the relation:

$$X_n = \mu + \begin{cases} \alpha_1(X_{n-1} - \mu) + e_n, & \text{if } X_{n-1} \leq d \\ \alpha_2(X_{n-1} - \mu) + e_n, & \text{if } X_{n-1} > d \end{cases}$$

The distinctive feature of some models from the threshold autoregressive class is the limit cycle behaviour. This makes the threshold autoregressive models suitable for the description of 'cyclic' phenomena.

In an extreme case we might set $\alpha_2 = 0$ for example. Then X_n follows an autoregressive process until it passes the threshold value d . At this point X_n returns to μ and the process effectively starts again. Thus we get cyclic behaviour as the process keeps resetting.

6.3 Random coefficient autoregressive models

Another modification of the AR class of models is that of autoregressive models for which the coefficient is random. In other words:

$$X_t = \mu + \alpha_t(X_{t-1} - \mu) + e_t$$

where $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$ is a sequence of independent random variables.

Such a model could be used to represent the behaviour of an investment fund, with $\mu = 0$ and $\alpha_t = 1 + i_t$ with i_t being the random rate of return.

The behaviour of these processes can vary widely, depending on the distribution chosen for the α_t , but is in general more irregular than that of the corresponding $AR(1)$.

6.4 Autoregressive models with conditional heteroscedasticity

Financial assets often display the following behaviour. After a large change in the asset price there follows a period of high volatility, which can be in either direction. Following small changes there tend to be further small changes. In other words, the variance of the process is dependent upon the size of the previous value. This is the property of *conditional heteroscedasticity*.

The words 'homoscedastic' and 'heteroscedastic' just mean having equal (*ie* constant) or different variances respectively. The 'c' is pronounced like a 'k' in these words.

The class of autoregressive models with conditional heteroscedasticity of order p – the $ARCH(p)$ – is defined by the relation:

$$X_t = \mu + e_t \sqrt{\alpha_0 + \sum_{k=1}^p \alpha_k (X_{t-k} - \mu)^2}$$

where e is a sequence of independent standard normal random variables. The simplest representative of the $ARCH(p)$ class is the $ARCH(1)$ model defined by the relation:

$$X_t = \mu + e_t \sqrt{\alpha_0 + \alpha_1 (X_{t-1} - \mu)^2}$$

If μ is zero, it can be shown that $\text{cov}(X_t, X_s) = 0$ for $s \neq t$ confirming that X_t is white noise with uncorrelated but not independent components.

The ARCH models have been used for modelling financial time series. If Z_t is the price of an asset at the end of the t -th trading day, it is found that the ARCH model can be used to model $X_t = \ln(Z_t/Z_{t-1})$, interpreted as the daily return on day t .

The ARCH family of models captures the feature frequently observed in asset price data that a significant change in the price of an asset is often followed by a period of high volatility. As may be seen from the $ARCH(1)$ model, a significant deviation of X_{t-1} from the mean μ gives rise to an increase in the conditional variance of X_t given X_{t-1} .

7 Appendix – Eigenvalues

λ is an *eigenvalue* of an $n \times n$ matrix A if there is a non-zero vector $\underline{x}$, such that:

$$A\underline{x} = \lambda\underline{x}$$

The vector $\underline{x}$ is known as the *eigenvector*. This equation is equivalent to $(A - \lambda I)\underline{x} = 0$ where I is the identity matrix, ie the matrix whose diagonal entries are 1 and whose off-diagonal entries are 0.

Hence we have a set of n linear equations in n unknowns, $\{x_1, \dots, x_n\}$. These equations have a non-zero solution, if and only if the matrix $A - \lambda I$ has zero determinant:

$$\det(A - \lambda I) = 0$$

The equation $\det(A - \lambda I) = 0$ can be solved for λ to find the eigenvalues.



Question

Calculate the eigenvalues of the matrix $A = \begin{pmatrix} 2 & 1 \\ 4 & 2 \end{pmatrix}$.

Solution

Here we have:

$$A - \lambda I = \begin{pmatrix} 2 & 1 \\ 4 & 2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 4 & 2 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} 2-\lambda & 1 \\ 4 & 2-\lambda \end{pmatrix}$$

We have to solve:

$$\det \begin{pmatrix} 2-\lambda & 1 \\ 4 & 2-\lambda \end{pmatrix} = 0$$

The determinant of the 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is $ad - bc$. So the eigenvalues of the matrix A are the solutions of the equation:

$$(2-\lambda)^2 - 4 = \lambda^2 - 4\lambda = \lambda(\lambda - 4) = 0$$

These are 0 and 4.

Chapter 14 Summary

Box-Jenkins methodology

The Box-Jenkins methodology gives us a way of fitting an $ARIMA(p, d, q)$ time series model to an actual data set. The method consists of the following steps:

- removing trends and cycles from the data set
- identifying a model from the $ARIMA(p, d, q)$ class
- estimating parameters
- diagnostic checks
- forecasting.

Removing trends

Time series data can be modelled efficiently only if stationary. In particular, any deterministic trends or cycles must be removed before applying the modelling procedure. There are various ways of doing this. In addition, a time series may still be non-stationary because it is integrated. In this case the time series must be differenced.

Let the set of observed values of the time series process be $\{x_t\}$.

Linear trends in the data can be removed by:

- least squares trend removal, ie we calculate $y_t = x_t - \hat{a} - \hat{b}t$ where $\hat{a}$ and $\hat{b}$ have been determined using linear regression
- differencing, ie we calculate $y_t = \nabla x_t = x_t - x_{t-1}$.

Seasonal trends in the data can be removed by:

- seasonal differencing, eg if seasonal variation is observed in monthly data, we calculate $y_t = x_t - x_{t-12}$
- method of moving averages (applying a filter), eg if seasonal variation is observed in monthly data, we calculate:

$$y_t = \frac{1}{12} (0.5x_{t-6} + x_{t-5} + \dots + x_{t-1} + x_t + x_{t+1} + \dots + x_{t+5} + 0.5x_{t+6})$$

- method of seasonal means, eg if seasonal variation is observed in monthly data, we subtract from each observation the estimated mean for that month.

It may be possible to remove other trends in the data via a transformation. For example, if the $\{x_t\}$ values appear to have an exponential trend, we could apply the transformation $y_t = \log x_t$.

Fitting an ARIMA process – choosing a value for d

The following principles can be used to choose an appropriate value for d .

1. If the sample autocorrelation function r_k decays slowly to 0, this indicates that there are still trends in the data and that the data should be differenced again.
2. Let $\hat{\sigma}_d^2$ denote the sample variance of the process $\nabla^d x_t$, then d can be set to the value which minimises $\hat{\sigma}_d^2$.

Fitting an ARIMA process – choosing values for p and q

If the underlying time series process is $MA(q)$, then we would expect the sample autocorrelation function $\hat{\rho}_k$ (or r_k) to cut off for $k > q$. It can also be shown that the estimator $\tilde{\rho}_k$ has the following approximate distribution for $k > q$:

$$\tilde{\rho}_k \doteq N\left(0, \frac{1}{n} \left(1 + 2 \sum_{i=1}^q \rho_i^2\right)\right)$$

We might conclude that the ACF cuts off for $k > q$ if 95% of the $\hat{\rho}_k$ values fall within the confidence interval:

$$\left[-1.96 \sqrt{\frac{1}{n} \left(1 + 2 \sum_{i=1}^q \hat{\rho}_i^2\right)}, +1.96 \sqrt{\frac{1}{n} \left(1 + 2 \sum_{i=1}^q \hat{\rho}_i^2\right)} \right]$$

If the underlying time series process is $AR(p)$, then we would expect the sample partial autocorrelation function $\hat{\phi}_k$ to cut off for $k > p$. It can also be shown that the estimator $\tilde{\phi}_k$ has the following approximate asymptotic distribution:

$$\tilde{\phi}_k \doteq N\left(0, \frac{1}{n}\right)$$

We might conclude that the PACF cuts off for $k > p$ if 95% of the $\hat{\phi}_k$ values fall within the confidence interval:

$$\left[-1.96 \sqrt{\frac{1}{n}}, +1.96 \sqrt{\frac{1}{n}} \right]$$

Otherwise, we look to fit an $ARMA(p, q)$ model. In practice, we might start with an $ARMA(1, 1)$ model and then apply diagnostic tests on the residuals to see whether this is a reasonable fit. If not, we would try adding more parameters.

Akaike's Information Criterion (AIC) states that we should only consider adding an extra parameter if this results in a reduction of the residual sum of squares by a factor of at least $e^{-2/n}$.

The formula for the $\hat{\rho}_k$ is given on page 40 of the *Tables*. The sample partial autocorrelation can then be calculated using the formulae (also on page 40 of the *Tables*) but with ρ_k replaced by $\hat{\rho}_k$.

Parameter estimation

Once we have identified p , d and q , we move forward with a time series of the form:

$$X_t = \alpha_1 X_{t-1} + \alpha_2 X_{t-2} + \cdots + \alpha_p X_{t-p} + e_t + \beta_1 e_{t-1} + \beta_2 e_{t-2} + \cdots + \beta_q e_{t-q}$$

The parameters, the alphas and betas can be estimated as follows:

- least squares estimation (which is equivalent to maximum likelihood estimation if the error terms can be assumed to be normally distributed)
- method of moments, where we equate population autocorrelations ρ_k with sample autocorrelations $\hat{\rho}_k$.

The final parameter of the model is σ^2 , the variance of the e_t , which may be estimated using:

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{t=p+1}^n \hat{e}_t^2 \text{ where } \hat{e}_t \text{ denotes the estimate of the residual at time } t$$

Diagnostic tests

If the model chosen is a good fit to the data, we would expect the estimates of the residuals $\{\hat{e}_t\}$ to show the characteristics of white noise (ie a set of uncorrelated random variables with zero mean). Examples of diagnostic tests include:

- checking that the graph of the $\{\hat{e}_t\}$ terms is patternless
- the $\{\hat{e}_t\}$ terms appear to be close to zero
- the turning point test: the number of points of inflexion in the graph of the $\{\hat{e}_t\}$ terms should fall within the 95% confidence interval:

$$\left[\frac{2}{3}(N-2) - 1.96\sqrt{\frac{16N-29}{90}}, \frac{2}{3}(N-2) + 1.96\sqrt{\frac{16N-29}{90}} \right]$$

- the sample autocorrelation $\hat{\rho}_k$ of the $\{\hat{e}_t\}$ terms is close to zero and has an approximate $N\left(0, \frac{1}{n}\right)$ distribution so that 95% of its values should fall within the confidence interval $\left[-1.96\sqrt{\frac{1}{n}}, +1.96\sqrt{\frac{1}{n}}\right]$
- the Ljung and Box 'portmanteau' test: if the $\{\hat{e}_t\}$ terms are white noise then they should be uncorrelated. Under the null hypothesis that the residuals are white noise, the sample autocorrelation r_k of the $\{\hat{e}_t\}$ terms satisfies:

$$n(n+2) \sum_{k=1}^m \frac{r_k^2}{n-k} \sim \chi_m^2 \text{ for each } m$$

This is a one-sided test.

Forecasting

Future values of the time series can be forecast using k -step ahead forecasting. We use the notation $\hat{x}_n(k)$ to be the estimate of the expected value of X_{n+k} (given the observations up to X_n). To determine $\hat{x}_n(k)$, we take our time series equation and:

- replace all unknown parameters by their estimated values
- replace the random variables $X_1, \dots, X_n$ by their observed values $x_1, \dots, x_n$
- replace the random variables $X_{n+1}, \dots, X_{n+k-1}$ by their forecast values $\hat{x}_n(1), \dots, \hat{x}_n(k-1)$
- replace the innovations $e_1, \dots, e_n$ by the residuals $\hat{e}_1, \dots, \hat{e}_n$
- replace the random variables $e_{n+1}, \dots, e_{n+k-1}$ by 0 (their expectations).

An alternative to 1-step ahead forecasting is exponential smoothing. We use the notation $\hat{x}_n(1)$ to be the estimate of the expected value of X_{n+1} (given the observations up to X_n).

$$\hat{x}_n(1) = \alpha \left[x_n + (1-\alpha)x_{n-1} + (1-\alpha)^2 x_{n-2} + \dots \right]$$

This is a weighted average of the past values but there is less emphasis on older values. The parameter α is called the smoothing parameter. Rearrangements include:

$$\hat{x}_n(1) = \alpha x_n + (1-\alpha)\hat{x}_{n-1}(1)$$

and:

$$\hat{x}_n(1) = \hat{x}_{n-1}(1) + \alpha [x_n - \hat{x}_{n-1}(1)]$$

Multivariate time series

We can write a univariate time series in multivariate (or vector) form.

For example, the time series $X_t = \alpha_1 X_{t-1} + \alpha_2 X_{t-2} + e_t + \beta e_{t-1}$ can be written as

$$\begin{pmatrix} X_t \\ X_{t-1} \\ X_{t-2} \end{pmatrix} = \begin{pmatrix} 0 & \alpha_1 & \alpha_2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} X_t \\ X_{t-1} \\ X_{t-2} \end{pmatrix} + \begin{pmatrix} 1 & \beta & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} e_t \\ e_{t-1} \\ e_{t-2} \end{pmatrix}$$

ie $\underline{X}_t = A\underline{X}_{t-1} + B\underline{e}_t$

The advantage of the vector form is that it displays the Markov property.

The vector process is stationary if the eigenvalues λ of the matrix A are all strictly less than 1 in magnitude. The eigenvalues are found by solving $\det(A - \lambda I) = 0$ where I is the identity matrix.

Cointegrated series

Two time series processes X and Y are called cointegrated if:

- (i) X and Y are $I(1)$ random processes
- (ii) there exists a non-zero vector (called the cointegrating vector) (α, β) such that $\alpha X + \beta Y$ is stationary.

We might expect that two processes are cointegrated if one of the processes is driving the other or if both are being driven by the same underlying process.

Other non-linear, non-stationary time series

Other examples of time series include:

- bilinear models, which exhibit 'bursty' behaviour:

$$X_n - \alpha(X_{n-1} - \mu) = \mu + e_n + \beta e_{n-1} + b(X_{n-1} - \mu)e_{n-1}$$

- threshold autoregressive models, which are used to model 'cyclical' behaviour:

$$X_n = \mu + \begin{cases} \alpha_1(X_{n-1} - \mu) + e_n, & \text{if } X_{n-1} \leq d \\ \alpha_2(X_{n-1} - \mu) + e_n, & \text{if } X_{n-1} > d \end{cases}$$

- random coefficient, autoregressive models:

$$X_t = \mu + \alpha_t(X_{t-1} - \mu) + e_t$$

where $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$ is a sequence of independent random variables.

- autoregressive conditional heteroscedasticity (ARCH) models, which are used to model asset prices, where we require the volatility to depend on the size of the previous value:

$$X_t = \mu + e_t \sqrt{\alpha_0 + \sum_{k=1}^p \alpha_k (X_{t-k} - \mu)^2}$$



Chapter 14 Practice Questions

- 14.1 The following table shows some data for the d th order differences of an observed time series x_t , $t = 1, 2, \dots, 100$:

Properties of $\nabla^d x_t$		$d = 0$	$d = 1$	$d = 2$	$d = 3$	$d = 4$
Sample autocorrelation coefficients	r_1	0.97	0.41	0.03	-0.24	-0.41
	r_2	0.92	-0.19	-0.49	-0.42	-0.25
	r_3	0.88	-0.17	-0.16	-0.01	0.06
	r_4	0.85	0.04	0.19	0.26	0.22
Sample variance		162.3	7.4	8.5	16.4	40.7

State, with reasons, the most appropriate value of d if this series is to be modelled using an $ARIMA(p, d, q)$ model.

- 14.2 A time series, X , is believed to conform to the $AR(1)$ model $X_n = \alpha X_{n-1} + \varepsilon_n$, where ε is a white noise process. The value of the parameter α is unknown.

- (i) The table below shows an extract of the calculation of the residuals for this model when α is assumed to equal 0.6 and 0.7.

n	...	15	16	17	18	19	20	...
x_n	...	371	507	B	449	272	76	...
Residuals ($\alpha = 0.6$)	...	357	A	-48	295	3	-87	...
Residuals ($\alpha = 0.7$)	...	354	247	-99	C	-42	-114	...

Complete the table by calculating the values of A , B and C .

- (ii) List the tests that could be applied to the residuals to test the model for goodness of fit.

- 14.3 An $ARIMA(p, d, q)$ model is to be fitted to the RPI data on page 152 of the *Tables*.

- (i) Explain why the value $d = 1$ would be considered the most appropriate choice for the parameter d .
- (ii) Comment on whether there is any evidence of seasonal variation in this dataset.

14.4 An $ARIMA(0,1,1)$ model of the form $\nabla E_t = \mu + \varepsilon_t + \beta \varepsilon_{t-1}$, where $\{\varepsilon_t\}$ is zero-mean white noise and $|\beta| < 1$, is to be fitted to the NAEI data on page 153 of the *Tables*.

- (i) State why the condition $|\beta| < 1$ has been imposed.
- (ii) Estimate the value of μ .
- (iii) Estimate the value of β by equating the sample and theoretical autocorrelations for lag 1 for the series $\{\nabla E_t\}$.
- (iv) Use this model to estimate E_{120} , the value of the series for Jan-02, given that $\hat{\varepsilon}_{119} = 3.1$.
- (v) Explain why this estimate might not be reliable.

14.5 From a sample of 50 consecutive observations from a stationary process, the table below gives values for the sample autocorrelation function (ACF) and the sample partial autocorrelation function (PACF):

Exam style

Lag	ACF	PACF
1	0.854	0.854
2	0.820	0.371
3	0.762	0.085

The sample variance of the observations is 1.253.

- (i) Suggest, giving reasons, an appropriate model based on this information. [2]
- (ii) Consider the $AR(1)$ model:

$$Y_t = a_1 Y_{t-1} + e_t$$

where e_t is a white noise error term with mean zero and variance σ^2 .

Calculate method of moments (Yule-Walker) estimates for the parameters of a_1 and σ^2 on the basis of the observed sample. [4]

- (iii) Consider the $AR(2)$ model:

$$Y_t = a_1 Y_{t-1} + a_2 Y_{t-2} + e_t$$

where e_t is a white noise error term with mean zero and variance σ^2 .

Calculate method of moments (Yule-Walker) estimates for the parameters of a_1 , a_2 and σ^2 on the basis of the observed sample. [7]

- (iv) List two statistical tests that could be applied to the residuals after fitting a model to time series data. [2]

[Total 15]

14.6 The following data is observed from $n = 500$ realisations from a time series:

Exam style

$$\sum_{i=1}^n x_i = 13,153.32, \quad \sum_{i=1}^n (x_i - \bar{x})^2 = 3,153.67 \quad \text{and} \quad \sum_{i=1}^{n-1} (x_i - \bar{x})(x_{i+1} - \bar{x}) = 2,176.03$$

- (i) Estimate, using the data above, the parameters μ , α_1 and σ from the model:

$$X_t - \mu = \alpha_1(X_{t-1} - \mu) + \varepsilon_t$$

where ε_t is a white noise process with variance σ^2 . [7]

- (ii) After fitting the model with the parameters found in (i), it was calculated that the number of turning points of the residuals series $\hat{\varepsilon}_t$ is 280.

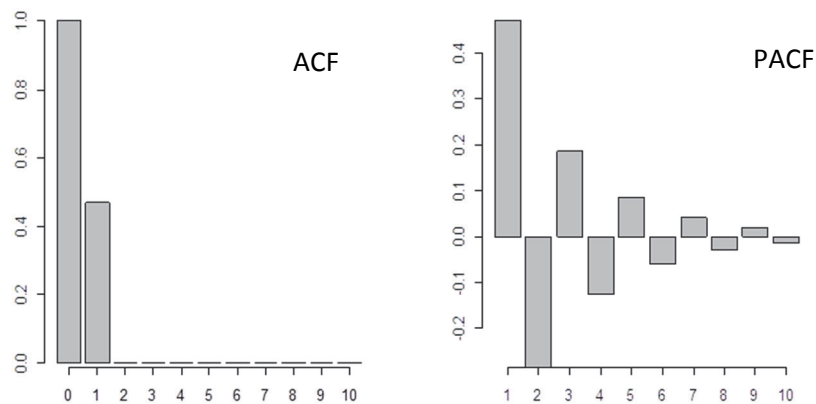
Perform a statistical test to check whether there is evidence that $\hat{\varepsilon}_t$ is not generated from a white noise process. [3]

[Total 10]

- 14.7 (i) State the three main stages in the Box-Jenkins approach to fitting an ARIMA time series model. [3]

Exam style

- (ii) Explain, with reasons, which ARIMA time series would fit the observed data in the charts below. [2]



Now consider the time series model given by:

$$X_t = \alpha_1 X_{t-1} + \alpha_2 X_{t-2} + \beta_1 e_{t-1} + e_t$$

where e_t is a white noise process with variance σ^2 .

- (iii) Derive the Yule-Walker equations for this model. [6]

- (iv) Explain whether the partial autocorrelation function for this model can ever give a zero value. [2]

[Total 13]

The solutions start on the next page so that you can
separate the questions and solutions.



Chapter 14 Solutions

- 14.1 The strongest clue in this question is the sample variance, which should have a small value when an appropriate value of d is found. We can rule out $d = 0$ because the sample variance is much bigger here. $d = 1$ or $d = 2$ seem likely candidates, since the sample variance is low for these, but then starts to increase when $d = 3$.

We also expect the sample autocorrelation coefficients to decay rapidly to zero. Again, this rules out $d = 0$. $d = 1$ looks good and has a more regular pattern than the higher values of d .

Putting these clues together suggests that we should select $d = 1$, as this value satisfies the required criteria, and going to $d = 2$ doesn't result in any worthwhile improvement.

In fact the series used here was a simulation of an ARIMA(2,1,0) process. So d really is equal to 1.

- 14.2 (i) **Completed table**

The completed table looks like this:

n	...	15	16	17	18	19	20	...
x_n	...	371	507	256	449	272	76	...
Residuals ($\alpha = 0.6$)	...	357	284	-48	295	3	-87	...
Residuals ($\alpha = 0.7$)	...	354	247	-99	270	-42	-114	...

The missing numbers are calculated as follows:

$$A = 507 - 0.6 \times 371 = 284$$

$$B - 0.6 \times 507 = -48 \Rightarrow B = 256$$

$$C = 449 - 0.7 \times 256 = 270$$

- (ii) **Tests on residuals**

The tests described in the Core Reading are:

- inspection of the graph of the residuals
- inspection of the SACF and SPACF
- the portmanteau test
- counting turning points.

14.3 (i) **Explain why $d = 1$**

The parameter d should be the smallest non-negative integer for which the series $\{\nabla^d Q_t\}$ can be considered to be stationary.

The graph in the *Tables* strongly suggests that $\{Q_t\}$ itself has an upward trend and is therefore not stationary.

The sample autocorrelations of $\{Q_t\}$ decay slowly from 1 which indicates differencing is required. However, the sample autocorrelations of $\{\nabla Q_t\}$ do not decay slowly from 1 which indicates that no further differencing is required.

The sample standard deviation of the values is minimised for $\{\nabla Q_t\}$. This also indicates that $d = 1$.

If there is any conflict between the two criteria then we should use the principle of parsimony in choosing the value for d .

(ii) **Seasonal variation**

Seasonal variation means that an annual cycle is present in the data.

The sample autocorrelation for lag 12 (ie 0.637) for the series $\{\nabla Q_t\}$, which we would otherwise consider to be stationary, is positive and significantly different from zero. This suggests the presence of a 12-monthly cycle.

Note that, because $\{Q_t\}$ doesn't appear to be stationary, the value of r_{12} for this series would be quite high, whether or not seasonality was present.

14.4 (i) **Reason for condition**

This condition ensures that the model is invertible.

(ii) **Estimate μ**

Since the white noise has mean zero, we know that $E[\nabla E_t] = \mu$.

If we equate $E[\nabla E_t]$ to the sample mean of the first differences shown in the *Tables*, this gives $\hat{\mu} = 0.4$.

(iii) **Estimate β**

We can calculate the autocovariances of $\{\nabla E_t\}$ for lags 0 and 1 for this model to be:

$$\gamma_0 = (1 + \beta^2)\sigma^2 \quad \text{and} \quad \gamma_1 = \beta\sigma^2$$

where σ^2 is the variance of the white noise series.

So:

$$\rho_1 = \frac{\beta}{1 + \beta^2}$$

Equating this to the sample value gives:

$$\frac{\beta}{1 + \beta^2} = -0.245$$

This leads to the quadratic equation:

$$0.245\beta^2 + \beta + 0.245 = 0 \Rightarrow \hat{\beta} = -0.262 \text{ or } -3.82$$

Since $|\hat{\beta}| < 1$, we conclude that $\hat{\beta} = -0.262$.

(iv) **Estimate E_{120}**

We first need to estimate ∇E_{120} . The actual value will be:

$$\nabla E_{120} = \mu + \varepsilon_{120} + \beta \varepsilon_{119}$$

We estimate this using the equation:

$$\nabla \hat{E}_{120} = \hat{\mu} + 0 + \hat{\beta} \hat{\varepsilon}_{119}, \text{ so } \nabla \hat{E}_{120} = 0.4 + 0 - 0.262(3.1) = -0.4$$

We can then calculate the required estimate as:

$$\hat{E}_{120} = E_{119} + \nabla \hat{E}_{120} = 134.1 + (-0.4) = 133.7$$

(v) **Why estimate may not be reliable**

We can see from the graph (and from the value of r_{12} for $\{\nabla E_t\}$) that the series contains a strong seasonal component, which we have not allowed for in this model.

The actual value for Jan-02 (not shown in the Tables) was 132.4, which is significantly different from our estimate.

14.5 This is Subject CT6, September 2008, Question 10.

(i) **Appropriate model**

From the figures given it looks like the ACF is decaying slowly and the PACF is cutting off after lag 2. This is a characteristic of an $AR(2)$ model. [2]

(ii) **Parameter estimates**

As a starter step:

$$\text{cov}(Y_t, e_t) = \text{cov}(a_1 Y_{t-1} + e_t, e_t) = \sigma^2 \quad [\frac{1}{2}]$$

Consider the autocovariance with a lag of 1:

$$\gamma_1 = \text{cov}(Y_t, Y_{t-1}) = \text{cov}(a_1 Y_{t-1} + e_t, Y_{t-1}) = a_1 \gamma_0 \Rightarrow \rho_1 = a_1 \quad [1]$$

Because we are told in the question that the sample ACF with lag 1 is 0.854, this is our estimate of ρ_1 , so we have:

$$0.854 = \hat{a}_1 \quad [\frac{1}{2}]$$

Consider the autocovariance with a lag of 0:

$$\gamma_0 = \text{cov}(Y_t, Y_t) = \text{cov}(a_1 Y_{t-1} + e_t, Y_t) = a_1 \gamma_1 + \sigma^2 \quad [\frac{1}{2}]$$

Because we are told in the question that the sample ACF with lag 1 is 0.854 (our estimate of ρ_1) and the sample variance is 1.253 (our estimate of γ_0), we have:

$$\begin{aligned} \hat{\gamma}_0 &= 1.253 \\ \frac{\hat{\gamma}_1}{\hat{\gamma}_0} &= 0.854 \Rightarrow \hat{\gamma}_1 = 0.854 \times 1.253 \end{aligned} \quad [1]$$

From $\gamma_0 = a_1 \gamma_1 + \sigma^2$:

$$1.253 = 0.854^2 \times 1.253 + \hat{\sigma}^2 \Rightarrow \hat{\sigma}^2 = 0.339 \quad [\frac{1}{2}]$$

(iii) **Parameter estimates**

Consider the autocovariance with a lag of 1:

$$\begin{aligned} \gamma_1 &= \text{cov}(Y_t, Y_{t-1}) = \text{cov}(a_1 Y_{t-1} + a_2 Y_{t-2} + e_t, Y_{t-1}) = a_1 \gamma_0 + a_2 \gamma_1 \\ \Rightarrow \rho_1 &= \frac{a_1}{1 - a_2} \end{aligned} \quad [1]$$

Consider the autocovariance with a lag of 2:

$$\begin{aligned} \gamma_2 &= \text{cov}(Y_t, Y_{t-2}) = \text{cov}(a_1 Y_{t-1} + a_2 Y_{t-2} + e_t, Y_{t-2}) = a_1 \gamma_1 + a_2 \gamma_0 \\ &= \frac{a_1^2}{1 - a_2} \gamma_0 + a_2 \gamma_0 = \frac{a_1^2 + (1 - a_2)a_2}{1 - a_2} \gamma_0 \end{aligned} \quad [1]$$

From this:

$$\rho_2 = \frac{a_1^2 + (1 - a_2)a_2}{1 - a_2} \quad [\frac{1}{2}]$$

Because we are told in the question that the sample ACF with lag 1 is 0.854 (our estimate for ρ_1) and that the sample ACF with lag 2 is 0.820 (our estimate of ρ_2), we have:

$$\begin{aligned} 0.854 &= \frac{\hat{a}_1}{1 - \hat{a}_2} \\ 0.820 &= \frac{\hat{a}_1^2 + (1 - \hat{a}_2)\hat{a}_2}{1 - \hat{a}_2} \end{aligned} \quad [1]$$

Replacing $\hat{a}_1$ by $0.854(1 - \hat{a}_2)$ in the second equation, we get:

$$\begin{aligned} 0.820 &= \frac{0.854^2(1 - \hat{a}_2)^2 + (1 - \hat{a}_2)\hat{a}_2}{1 - \hat{a}_2} = 0.854^2(1 - \hat{a}_2) + \hat{a}_2 \\ \Rightarrow 0.820 - 0.854^2 &= \hat{a}_2(1 - 0.854^2) \\ \Rightarrow \hat{a}_2 &= 0.335 \end{aligned} \quad [1]$$

By substituting this back into the first equation above, we get $\hat{a}_1 = 0.568$. [½]

Consider the autocovariance with a lag of 0:

$$\gamma_0 = \text{cov}(Y_t, Y_t) = \text{cov}(a_1 Y_{t-1} + a_2 Y_{t-2} + e_t, Y_t) = a_1 \gamma_1 + a_2 \gamma_2 + \sigma^2 \quad [½]$$

Because we are told in the question that the sample ACF with lag 1 is 0.854, the sample ACF with lag 2 is 0.820 and the sample variance is 1.253, we have:

$$\begin{aligned} \hat{\gamma}_0 &= 1.253 \\ \frac{\hat{\gamma}_1}{\hat{\gamma}_0} &= 0.854 \Rightarrow \hat{\gamma}_1 = 0.854 \times 1.253 \\ \frac{\hat{\gamma}_2}{\hat{\gamma}_0} &= 0.820 \Rightarrow \hat{\gamma}_2 = 0.820 \times 1.253 \end{aligned} \quad [1]$$

From $\gamma_0 = a_1 \gamma_1 + a_2 \gamma_2 + \sigma^2$:

$$\begin{aligned} 1.253 &= 0.568 \times 0.854 \times 1.253 + 0.335 \times 0.820 \times 1.253 + \hat{\sigma}^2 \\ \Rightarrow \hat{\sigma}^2 &= 0.301 \end{aligned} \quad [½]$$

(iv) **Tests**

We could mention any two of the following:

- Portmanteau (Ljung and Box) test
- Turning points test
- Inspection of the graph of the residuals

- Inspection of the values of the sample autocorrelation function based on their 95% confidence intervals
- Inspection of the values of the sample partial autocorrelation function based on their 95% confidence intervals [2]

14.6 This is Subject CT6, September 2009, Question 6.

(i) **Estimation of parameters**

The value of μ is estimated by the sample mean:

$$\hat{\mu} = \bar{x} = \frac{13,153.32}{500} = 26.31 \quad [1]$$

To estimate α_1 and σ , we consider the Yule-Walker equations:

$$\begin{aligned} \gamma_0 &= \text{var}(X_t) = \text{cov}(X_t, X_t) \\ &= \text{cov}(\alpha_1 X_{t-1} + \varepsilon_t, X_t) \\ &= \alpha_1 \text{cov}(X_{t-1}, X_t) + \text{cov}(\varepsilon_t, X_t) \\ &= \alpha_1 \gamma_1 + \sigma^2 \end{aligned} \quad [1]$$

since:

$$\text{cov}(\varepsilon_t, X_t) = \text{cov}(\varepsilon_t, \alpha_1 X_{t-1} + \varepsilon_t) = \alpha_1 \text{cov}(\varepsilon_t, X_{t-1}) + \text{cov}(\varepsilon_t, \varepsilon_t) = 0 + \sigma^2$$

In addition:

$$\begin{aligned} \gamma_1 &= \text{cov}(X_t, X_{t-1}) \\ &= \text{cov}(\alpha_1 X_{t-1} + \varepsilon_t, X_{t-1}) \\ &= \alpha_1 \text{cov}(X_{t-1}, X_{t-1}) + \text{cov}(\varepsilon_t, X_{t-1}) \\ &= \alpha_1 \gamma_0 + 0 \end{aligned} \quad [1]$$

$$\Rightarrow \rho_1 = \frac{\gamma_1}{\gamma_0} = \alpha_1 \quad [\frac{1}{2}]$$

Now using the formulae for $\hat{\gamma}_k$ and $\hat{\rho}_k$ given on page 40 of the *Tables*:

$$\hat{\gamma}_0 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{3,153.67}{500} \quad [\frac{1}{2}]$$

$$\hat{\gamma}_1 = \frac{1}{n} \sum_{i=1}^{n-1} (x_i - \bar{x})(x_{i+1} - \bar{x}) = \frac{2,176.03}{500} \quad [\frac{1}{2}]$$

So:

$$\hat{\rho}_1 = \frac{\hat{\gamma}_1}{\hat{\gamma}_0} = \frac{2,176.03}{3,153.67} = 0.6900 \quad [1/2]$$

$$\hat{\alpha}_1 = \hat{\rho}_1 = 0.6900 \quad [1]$$

and:

$$\begin{aligned} \hat{\gamma}_0 &= \hat{\alpha}_1 \hat{\gamma}_1 + \hat{\sigma}^2 \\ \Rightarrow \frac{3,153.67}{500} &= 0.6900 \times \frac{2,176.03}{500} + \hat{\sigma}^2 \\ \Rightarrow \hat{\sigma}^2 &= \frac{3,153.67}{500} - 0.6900 \times \frac{2,176.03}{500} = 3.3044 \\ \Rightarrow \hat{\sigma} &= 1.8178 \end{aligned} \quad [1]$$

(ii) **Turning point test**

The null and alternative hypotheses are:

H_0 : the residuals are from a white noise process

H_1 : the residuals are not from a white noise process [1]

Using the formulae from page 42 of the *Tables*, we have:

$$\begin{aligned} E(T) &= \frac{2}{3} \times 498 = 332 \\ \text{var}(T) &= \frac{16 \times 500 - 29}{90} = 88.567 \end{aligned}$$

The value of the test statistic is:

$$\frac{280 + 0.5 - 332}{\sqrt{88.567}} = -5.47 \quad [1]$$

which should be from a $N(0,1)$ distribution if H_0 holds. Since $-5.47 < -1.96$ we have very strong evidence to reject H_0 . This suggests that the residuals are not from a white noise process. [1]

14.7 This is Subject CT6, September 2013, Question 9.

(i) **Box-Jenkins approach**

The three main stages in the Box-Jenkins methodology are:

- Tentative identification of a model from the ARIMA class. [1]
- Estimation of parameters in the identified model. [1]
- Diagnostic checks. [1]

(ii) **ARIMA time series to fit the observed data in the charts**

The ACF cuts off (becomes 0) at all lags greater than 1, whereas the PACF decays towards 0. Hence we have an $MA(1)$.

[2]

(iii) **Yule-Walker equations**

We start with some useful preliminary equations:

$$\begin{aligned}\text{cov}(X_t, e_t) &= \text{cov}(\alpha_1 X_{t-1} + \alpha_2 X_{t-2} + \beta_1 e_{t-1} + e_t, e_t) \\ &= \alpha_1 \text{cov}(X_{t-1}, e_t) + \alpha_2 \text{cov}(X_{t-2}, e_t) + \beta_1 \text{cov}(e_{t-1}, e_t) + \text{cov}(e_t, e_t) \\ &= 0 + 0 + 0 + \sigma^2 = \sigma^2\end{aligned}$$

[1]

and:

$$\begin{aligned}\text{cov}(X_t, e_{t-1}) &= \text{cov}(\alpha_1 X_{t-1} + \alpha_2 X_{t-2} + \beta_1 e_{t-1} + e_t, e_{t-1}) \\ &= \alpha_1 \text{cov}(X_{t-1}, e_{t-1}) + \alpha_2 \text{cov}(X_{t-2}, e_{t-1}) + \beta_1 \text{cov}(e_{t-1}, e_{t-1}) + \text{cov}(e_t, e_{t-1}) \\ &= \alpha_1 \text{cov}(X_t, e_t) + 0 + \beta_1 \text{var}(e_t) + 0 \\ &= \alpha_1 \sigma^2 + \beta_1 \sigma^2 = (\alpha_1 + \beta_1) \sigma^2\end{aligned}$$

[1]

The Yule-Walker equations are as follows:

$$\begin{aligned}\gamma_0 &= \text{cov}(X_t, X_t) \\ &= \text{cov}(\alpha_1 X_{t-1} + \alpha_2 X_{t-2} + \beta_1 e_{t-1} + e_t, X_t) \\ &= \alpha_1 \text{cov}(X_{t-1}, X_t) + \alpha_2 \text{cov}(X_{t-2}, X_t) + \beta_1 \text{cov}(e_{t-1}, X_t) + \text{cov}(e_t, X_t) \\ &= \alpha_1 \gamma_1 + \alpha_2 \gamma_2 + \beta_1 (\alpha_1 + \beta_1) \sigma^2 + \sigma^2\end{aligned}$$

[1]

$$\begin{aligned}\gamma_1 &= \text{cov}(X_t, X_{t-1}) \\ &= \text{cov}(\alpha_1 X_{t-1} + \alpha_2 X_{t-2} + \beta_1 e_{t-1} + e_t, X_{t-1}) \\ &= \alpha_1 \text{cov}(X_{t-1}, X_{t-1}) + \alpha_2 \text{cov}(X_{t-2}, X_{t-1}) + \beta_1 \text{cov}(e_{t-1}, X_{t-1}) + \text{cov}(e_t, X_{t-1}) \\ &= \alpha_1 \gamma_0 + \alpha_2 \gamma_1 + \beta_1 \sigma^2\end{aligned}$$

[1]

$$\begin{aligned}\gamma_2 &= \text{cov}(X_t, X_{t-2}) \\ &= \text{cov}(\alpha_1 X_{t-1} + \alpha_2 X_{t-2} + \beta_1 e_{t-1} + e_t, X_{t-2}) \\ &= \alpha_1 \text{cov}(X_{t-1}, X_{t-2}) + \alpha_2 \text{cov}(X_{t-2}, X_{t-2}) + \beta_1 \text{cov}(e_{t-1}, X_{t-2}) + \text{cov}(e_t, X_{t-2}) \\ &= \alpha_1 \gamma_1 + \alpha_2 \gamma_0\end{aligned}$$

[1]

In general, for lags $k \geq 2$:

$$\gamma_k = \alpha_1 \gamma_{k-1} + \alpha_2 \gamma_{k-2}$$

[1]

(iv) ***Can the partial auto-correlation function ever give a zero value?***

For an $MA(q)$ process, where $q \geq 1$, the PACF tends towards 0 but does not completely cut off.

[1]

Here, we have an $ARMA(2,1)$ process, ie $q \geq 1$. Hence the PACF tends towards 0 but does not completely cut off. There will always be a small partial autocorrelation.

[1]

All study material produced by ActEd is copyright and is sold for the exclusive use of the purchaser. The copyright is owned by Institute and Faculty Education Limited, a subsidiary of the Institute and Faculty of Actuaries.

Unless prior authority is granted by ActEd, you may not hire out, lend, give out, sell, store or transmit electronically or photocopy any part of the study material.

You must take care of your study material to ensure that it is not used or copied by anybody else.

Legal action will be taken if these terms are infringed. In addition, we may seek to take disciplinary action through the profession or through your employer.

These conditions remain in force after you have finished using the course.